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MINISTRY OF AGRICULTURE



Practical Guideline for

Application of Reproductive Biotechnology in Dairy Cattle

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Practical Guideline for Application of Reproductive Biotechnology in Dairy Cattle

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Table of contents

Page

Lists of tables.....i

Lists of figures.....ii

Preface.....iii

Acronym.....iv

1. Introduction.....1

2. Objective.....3

3. Application of Reproductive biotechnology for genetic improvement and breed multiplication.....3

3.1. Guiding principles for Implementation of AI and Pregnancy Diagnosis.....3

3.1.1 Content of the training.....4

3.1.2 Inputs and supplies for AI service delivery.....5

3.1.3 Selection of the breeding cow/heifer for Artificial Insemination.....8

3.1.4 Artificial Insemination of Dairy Cattle.....9

3.1.5 Pregnancy Diagnosis.....10

3.1.6 Determining the Stages of Pregnancy.....12

3.1.7 Record keeping during Artificial Insemination.....15

3.2 Estrus synchronization in breeding cows/heifers.....16

3.2.1 Animal handling facilities and Personal Protection Equipment.....16

3.2.2 Selection of indigenous cows and heifers for synchronization.....16

3.2.3 Estrus Synchronization protocols.....17

3.2.4 Detection of Estrus and optimal time for insemination.....18

3.2.5 Time of insemination.....19

3.3 Application of Embryo transfer technology in dairy animals.....20

3.3.1. Preliminary preparation.....20

3.3.2 Preparation of cows for embryo production and transfer.....22

3.3.3 Embryo production and transfer.....23

4. Reproductive Health management and Biosecurity measures.....30

4.1 Reproductive health management of breeding animals.....30

4.1.1 Regular Reproductive health Inspection and Corrective measures.....31

4.1.2 Prevention and case management of repeat breeder.....32

4.1.3 Biosecurity measures.....32

4.1.4 Management of vaginal and uterine prolapse in breeding cows.....33

4.2 Management of hypocalcaemia in breeding cows.....34

4.3 Management of mastitis in breeding cows.....34

4.4 Common infectious and parasitic diseases.....34

4.5 Calves Health management....38
4.6 Animal Health Records....38
5. Breeding of dairy cattle....39
5.1. Indigenous cattle breed improvement for Mixed crop-livestock production system....39
5.2 Indigenous cattle breed improvement for Pastoral production system....40
5.3 Urban and peri-urban dairy....40
5.4 Record keeping....44
6. Husbandry of breeding animals....44
6.1 Types of animal feed resources....44
6.2 Feed and feeding management of breeding dairy cattle (Feeding regime for different age group indicated under annex 4)....45
6.3. Feeding management of heifers....46
6.4 Feeding management of calves and immature heifers....46
6.5 Feeding silage-based diets to dairy cattle....47
6.6. Dairy animal housing and housing requirements....47
6.7 Calf management....49
6.7.1 Care of the newborn....49
6.7.2 Colostrum feeding (1-4 days)....50
6.7.3 Housing the newborn calf....50
6.7.4 Health care for calves....51
7. Dairy Farm Recording....51
References....57
Annex 1....59
Procedures and steps in dairy cattle AI....59
Annex 2....62
Sub clinical mastitis screening methods and test procedures....62
1. Screening Test Procedures....62
2. Mastitis Prevention and Control....62
Annex 3....63
Procedures and steps to re-place the prolapsed uterus....63
Annex 4....64
Neonate calf, pre weaning calf, growing heifers, heifers and breeding cow/heifers feeding regime....64
Colostrum feeding (1-4 days)....64

Lists of Tables

Page

Table 1. Materials and equipment to be available for semen transportation and storage, and efficient AI service.....6
Table 2. Key indicators for bovine pregnancy diagnosis at different pregnancy stages..11
Table 3. Selection of individual donors and recipients for embryo production and transfer.....18
Table 4. Materials required for embryo retrieval.....20
Table 5. Items for embryo handling, evaluation and grading.....21
Table 6. Stages of embryonic development with respective descriptions.....22
Table 7. Embryo quality grade assessment guide.....24
Table 8: Common types of vaccines and calendar of vaccination.....29
Table 9: Implementation of deworming and spraying strategies in the central highland.....30
Table 10: Regular testing of animals for early disease diagnosis and screening.....31
Table 11. AI data recording sheet.....45
Table 12: Animal health observation and treatment record.....46
Table 13: Vaccination record.....46
Table 14. Mortality record.....46
Table15. Amount of colostrum and milk to be fed to growing calves.....55
Table 16. Example of a feeding schedule for calves.....57
Table 17. Partial replacement of milk for whey for early weaning of a calf.....57
Table 18. Preparation of supplemental concentrate mix for young growing heifers after weaning.....58
Table 19. Daily concentrate allowance (post-weaning feeding management to calving).....59
Table 20. Concentrate diet formulated for lactating cows under local conditions.....59

Lists of figures

Page

Figure 1. The O-ring on the insemination gun used to fix insemination sheath.....10

Figure 2. Decision for the best breeding time in cattle.....17

Figure 3. Donor Superovulation and Embryo recovery protocol.....19

Figure 4. Level of Foley catheter insertion in the uterine horn for embryo retrieval...21

Figure 5. Schematic explanation for two tiers using indigenous cattle genetic improvement scheme 32

Figure 6. Mating plan suggested for medium to high input-output milk shed areas delineated for crossbreeding.....34

Figure 7. Mating plan suggested for low input-output milk shed areas delineated for cross breeding.....35

Figure 9. A zero grazing/conventional barn complete with a sun shade structure.....40

Figure 10. Individual calf pen for neonatal calves.....42

Figure 11. a, b, c. Simulation of left hand manipulation cervix during Recto vaginal insemination.....52

Figure 12. a, b, c Simulation of left hand manipulation cervis to insert insemination gun during Recto vaginal insemination.....52

Figure 13. Simulation for the correct (a) and incorrect semen deposition site in the uterus during (b,c) Recto vaginal insemination.....52

Figure 14. Screening test methods for subclinical mastitis a. Strip cup test; b. CMT..53

Figure 15. Dipping techniques after milking.....53

Preface

Reproductive technologies, since inception in 1940s, had advanced with the widespread establishment of artificial insemination and the introduction of frozen semen in cattle breeding, and impacted the dairy industry in the world. In the early 1960s, progestogens administered by the intravaginal route were shown to be highly effective techniques for the control of the oestrous cycle and breeding season in cattle, sheep and goats. In the 1970s the availability of prostaglandins (PGs) and gonadotrophin releasing hormone (GnRH) further widened the possibilities for oestrus control in cattle. These technologies have enabled farmers to improve the biological efficiency of their livestock. Quite apart from the possible contribution in enhancing the selection intensity and reduction of generation interval, knowledge-based application of reproductive technologies provided options for the control and manipulation of reproductive function and traits of interest.

Artificial insemination, the most widely used reproductive technology, and estrus synchronization have a long history in Ethiopia. However, these long-term practices have brought about insignificant genetic improvement, but clearly showed that milk production can be improved by about more than three folds in crossbred cattle. This accounts for inconsistent and inefficient use of assisted reproductive technologies (AI, estrus synchronization) that have revolutionized the dairy industry in the world. To change the previous practices the FDRE Ministry of Agriculture has developed a 10-year dairy development strategy whereby the key gaps and set of priority interventions were identified. The designed interventions to produce improved dairy animals include selecting the appropriate genotype and production environment, putting in place key facilities, upholding a sound reproductive, feeding and health management, and husbandry of breeding animals. Of all the gaps identified limited technical knowledge

and skill were among the most important problems constrained the key procedures, protocols and requirements to implement these techniques with greater efficiency at small- and large-scale. As the control over the function of reproduction is technically more complex, clearly the skills required by those working with them must be updated. Professionals need to have an extensive knowledge of issues associated with animal care and welfare integrated into livestock production systems to ensure intended success in reproductive performance. It should also be emphasized that the provision of optimal nutritional management, maximizing animal comfort, minimizing stress, and maintaining good animal health are all prerequisites for the successful application of reproductive technologies. Therefore, this document guides practical application and implementation of reproductive biotechnology tools and enhances the knowhow and skill set of experts, technicians and dairy producers engaged in dairy production to exploit untapped potential and increase contribution of the dairy sector for food and nutrition security, employment creation and overall economic development of the nation.



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Acronym

AI	Artificial Insemination
AITs	Artificial Insemination Technicians
BCS	Body Condition Score
CBPP	Contagious bovine pleuropneumonia
CB	Crossbreeding
CIDR	Controlled Internal Drug Release, progesterone source
Cow ID	Cow Identification
BVDD	Bovine Viral Diarrhoea Disease
CL	Corpus Luteum
CMT	California Mastitis Test
CSA	Centre Statistical Agency
DCAD	Dietary Cation-Anion Difference
EBV	Estimated Breeding Value
EIAR	Ethiopia International Agriculture Research
FMD	Foot and Mouth Disease
FSH	Follicle Stimulating Hormone
HF	Holstein Friesian
IBR	Infectious Bovine Rhinotracheitis
IV	Intramuscular Vein
L/day	Litters per a day
L	Litter
LDI	Livestock Development Institute
LSD	Lumpy Skin Disease
MM	Milli meter
NAGI	National Artificial Genetics Institute
PD	Pregnancy Diagnosis
PGF	Placental Growth Factor
PPE	Persona Protocol Equipment

1. Introduction

Ethiopia possesses more than 70 million cattle, 52.5 million goats and 8.1 million camels; that make the country to bestow with the largest livestock population in Africa mainly constituted of indigenous genetic resources with their respective contribution to total national annual milk output: cattle (81.2%), camels (15%), and goats (3.8%) (CSA, 2021).

The Ministry of Agriculture with its affiliated institutions such as NAGI/LDI and EIAR have been engaged in a long-term dairy animals genetic improvement endeavor via semen production and cross-breeding of indigenous cattle breeds. The purpose of crossbreeding particularly cattle with high producing exotic temperate breeds (HF and Jersey) and was to combine their high milk yield potential with the adaptive potential of local breeds (Seyoum et. al., 2022). The results of this long-term engagement have clearly showed that milk production can be improved by about three to seven-folds in cattle. Smallholder farmers who possess crossbred dairy cattle could generate substantially higher income (40%) than those who keep only indigenous cattle (Agajie et al., 2016).

Despite subsequent efforts that have been made by government and non-governmental institution and huge investment used up in the last 60 years, the proportion of crossbred dairy cattle in the country is less than 3% of the national cattle population. Application of reproductive biotechnology (Artificial insemination, estrous synchronization, embryo transfer technologies) can help to increase crossbred cattle population and superior germplasm.

Artificial insemination which is the most widely used reproductive biotechnology and estrus synchronization have a fairly long history in Ethiopia. They have been practiced

to enhance genetic progress and hasten multiplication of improved dairy heifers at small scale and large-scale production systems. These technologies in conjunction with sexed semen and the introduction of embryo transfer technologies would speed up genetic progress and increase the number of animals that can be bred from global and local elite parents.

The pillars of interventions to produce improved dairy animals include selecting the appropriate genotype and production environment, putting in place key facilities, upholding a sound reproductive, feeding and health management (including biosecurity measures), proper record keeping, and husbandry of breeding animals. Practical implementation of reproductive herd health and associated management interventions can help to upgrade reproductive health and immunity of breeding animals. Accordingly, the EFDR ministry of agriculture and fishery has developed dairy development strategy 10 year and current development initiative (Lemat) where by the gap and intervention strategies were identified. And hence, the gap in technical knowledge and skill is among the most important problem constraining understanding of the key procedures, protocols and requirements to implement these techniques with greater efficiency at small- and large-scale dairy production systems. Furthermore, poor reproductive health management, limited biosecurity measures, poor diagnostic techniques plus lack of selection breeding of indigenous breeds or target cross breeding are the major ones whereby developing and upgrading the skill is prominent intervention area. Thus, this document guides practical implementation of reproductive biotechnology tools and enhances the knowhow and skill for experts, technicians and dairy producers engaged in dairy production to exploit untapped potential and increase the contribution of dairy sector for food security and overall economic development of the nation.

2. General objective

To enhance the transfer of knowledge, skill and attitudes to AI Technician, experts and dairy producers to support breed improvement endeavors and increase productivity of dairy animals for sustainable food and nutritional security in the country

Specific objectives

- To guide and assist trainers to deliver training on artificial insemination, estrus synchronization management and embryo transfer technologies, pregnancy diagnosis and associated processes
- To enhance knowledge of dairy farmers on artificial insemination and estrus synchronization management
- To improve semen and embryos processing and dissemination mechanisms for dairy animals breed improvement and multiplication
- To introduce standardized procedures of reproductive technology tools and utilization of related equipment.
- To promote best practices and experiences on dairy cattle husbandry and reproductive health management
- To enable farmers to practice record keeping in the dairy farms

3. Application of Reproductive biotechnology for genetic improvement and breed multiplication

3.1 Guiding principles for Implementation of AI and Pregnancy Diagnosis

Though various factors contribute for inefficient AI, lack of skill and devotion of AITs take the cornerstone for this inefficiency while performing effective AI technique, estrus synchronization and pregnancy diagnosis. Likewise, as it is well noted through various mechanisms, some technical and skill gaps have been observed in most of the AI Technicians in the country. Therefore, it is mandatory that AI Technicians should partake in a short-term refreshment training after every two years of service on specified

training courses listed below.

3.1.1 Content of the training

1. Reproductive anatomy and physiology of dairy cattle. This will help them

    - To understand the anatomy, topography and functions of reproductive organs of breeding animals
    - To recognize morphological, physiological and behavioral changes that occur during estrus period, and
    - To recognize the palpable changes of reproductive organs during estrus period through rectal palpation.

2. Semen handling, storage and artificial insemination techniques. The training should address areas of failure to stick to standards in equipment handling, insemination procedures and appropriate sanitations.
3. Selection and appraisal of breeding animals. The AITs' need to be trained on how:

    - To select cows/heifers based on set dairy characters for breeding objectives/ purpose
    - To undertake appraisal on the fitness of body condition of cows/heifers for breeding
    - To determine age and physiological status of cows/heifers to be bred

4. Early Pregnancy Diagnosis. AITs should acquire the skill to diagnose pregnancy up to two months post insemination and be able to advice the producers to care for pregnant cows/heifers to minimize reproductive loses.
5. Record keeping. The AITs must be able to take records of all activities related to the AI service delivered and report the record/data to the report managing body. Thus, AITs must understand the critical role and purpose of keeping in the genetic improvement program, enhancement of individual dairy cattle or herd production and productivity. The content of the record must not preclude producers' profile,

animal history and individual feature of the breeding animal such as Cow ID, Breed type, blood level, history of past AI or parturition, frequency/repetition of insemination, profile of Bull used, PD test result and calve gender, etc.), and application of technical formulae in measuring reproductive efficiencies/indexes is indispensable.

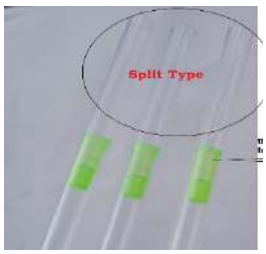
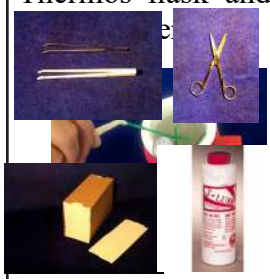
3.1.2 Inputs and supplies for AI service delivery

Use of standardized inputs (materials, tools and equipment; Table 1) for the effective and efficient AI service delivery is mandatory, and AITs and service givers have to stick to the recommended standards.

Table 1. Materials and equipment to be available for semen transportation and storage, and efficient AI service

Input/Supply item	Description
Liquid nitrogen container 	2 or 3L capacity for routine service delivery • at least 1.8 -2.8L, • a neck diameter of 30 - 51 mm, • a static evaporation loss rate of <0.1L/day, • at least 3 stainless teal canisters and 2 more replacements, • if the canisters are sieving type at the bottom, the canisters should have 5 goblets, • <12 cm Canister height with single level, • >100 straws (0.25ml size) capacity, • Must have a liquid nitrogen level measuring rod

Dewar (Medium size) 	**35L holding capacity used for semen storage at district level** • Need to have a minimum of 33L holding capacity with bottom pad or extension for footing and mobility/portable • A neck diameter of 51 to 70 mm • A static evaporation loss rate not > 0.15L/day • Should have at least 6 stainless steel canisters and 2 more replacements • At least 12 Goblets as per the model of the dewar and 4 more replacements • Have a canister height of 24 cm for double level • Must also have liquid nitrogen level measuring rod • It should have a straw (0.25ml) holding capacity >3000 in its 6 canister**s**
Dewar (Large size) 	**40L capacity used for semen storage and transportation by stations and sub stations** • It should be with double level goblet holding canister with protective rubber rings and bottom pad • The evaporation rate should be as low as possible with a static evaporation loss rate < 0.21 L/day • A neck diameter < 70 mm • It should have a static holding time not less than 190 days • Must have at least seven stainless steel canisters (including one for replacement) with a holding capacity of > 8400 straws (0.25ml) in its double level
Dewar (Large size) 	**47 liters capacity used for semen storage and transportation by stations and sub stations** • It should have a static evaporation loss rate of < 0.38 L/day • A neck diameter < 120 mm • A static holding time > 124 days • It should have at least eleven stainless steel canisters (including one for replacement) with a holding capacity of > 24000 straws (0.25ml straw) in its double level

Artificial insemination gun 	• Must be made of high-quality stainless-steel material with a barrel length of 435 mm and an outer diameter of 3.60 mm which is compatible with sheath that has outer diameter of 4.50 mm for a 0.25 ml straw • The plunger length should be 449 mm • Its barrel should be attached with a conical-shape stainless steel adopter along with a Plastic O-Ring. Preferably it shall be conical O-ring with auto-lock style • Must have a Gun Container
AI sheath 	• AI sheath must be made from non-toxic material (please observe manufacturer specification) • The length should be 443 mm with an outer diameter of 4.50 mm which is compatible with 0.25ml straw • It should have an adapter to fix the straw at the center to guarantee expulsion of all the semen without back-flow into the gun chamber/sheath • Preferably the sheath is individually packed and split end type for use by both conical O-ring and fixer type AI gun
Glove 	• It should be made of low-density polyethylene material with no tears or blisters or any nicks and cuts along the edges • It should have a thickness of 25µm (double wall thickness of 0.05 mm) • It should have a length of 920-930 mm
Thermos flask and 	• For accurate measurement of thawing temperature during AI • The thermometer should be calibrated and annually certified to fulfill LDI set standards
Paper towel and Lubricant	• Materials used for drawing straws from dewar, for cutting straw end, for drying the straws and lubricant for transrectal access need to be ready before one starts the AI service

3.1.3 Selection of the breeding cow/heifer for Artificial Insemination

Body CONDITION at calving and at the subsequent insemination influence the interval from calving to first estrus and also conception rate. Thus,

- Breeding cows should have optimal condition score between 3.0 and 4.0 (on 1-5 scale). Cows/heifers that are too thin (< 2.5 BCS) that are losing condition will have delayed estrus and poor conception rates.
- Cows that are too fat (>4 BCS) at calving are likely to have calving difficulties and are more prone to early fetal death.
- Cows/heifers to be selected for AI or synchronization should be conditioned starting two months before breeding
- The cow should be in good health. It should be free from any infection of the reproductive tract; particular attention should be paid to cows that had abnormal calving.

Other parameters that need to be considered during selection of cows and heifers for reproductive biotechnology:

- Cows should rest for at least 60 days after calving before they are inseminated again as they need sufficient time for involution of the uterus and to return to normal fertility
- The AI technician must make sure that the cow is well into estrus and not pregnant.
- Farmers should keep cows in estrus or injected with hormones away from bulls
- It is advisable to cull or castrate local bulls that can interfere with the breed improvement program
- Crossbred bulls SHOULD NOT be used to inseminate indigenous cows/heifers

3.1.4 Artificial Insemination of Dairy Cattle

a. Preparations for Insemination (Pre-Insemination Checks)

- Restrain the cow before thawing the straws for insemination. The restraint area should be free of stressful conditions (Unnecessary excitement may interfere with physiological mechanisms important to achieving a good conception rate).
- Confirm identity of the cow/heifer to be inseminated.
- Make a quick appraisal of the animal submitted for AI for presence of correct estrus signs, good body condition and physical stature, and freedom from disease/disorder.
- Always bear in mind that selecting the appropriate location for AI requires consideration of safety for both the animal and technician, ease of use and protection from adverse conditions including bad weather.

b. Taking Hygienic and safety measures

- Apply good sanitary procedures and practices throughout the insemination process.
- Assemble all insemination devices, protect from contamination and maintain them at optimal temperature.
- Avoid using products that are irritant and spermicidal lubricants.
- The perineum must be thoroughly wiped clean with a paper towel

c. Semen Thawing and Loading Procedure

Follow appropriate thawing and loading procedures as indicated in Annex 1.

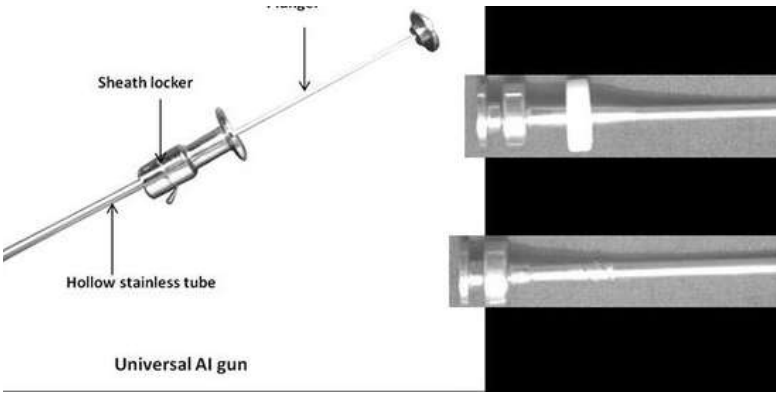


Fig 1. The O-ring on the insemination gun used to fix insemination sheath

d. Recto-vaginal Insemination Technique

Regardless of whether you are left or right-handed, it is recommended that you use your left hand in the rectum to guide the insemination gun through the reproductive tract. This is because the rumen of the cow lies on the left side of the abdominal cavity, displacing the reproductive tract slightly to the right. Thus, you will find it much easier to locate and manipulate the tract with your left as opposed to right hand. For the detail insemination procedures and steps follow the insemination technique as indicated in the annex 1.

3.1.5 Pregnancy Diagnosis

An accurate determination of pregnancy requires a good understanding of the anatomy and topography of the reproductive organs as well as the functional changes at different stages of pregnancy. Failure to do so or wrong manipulation of early pregnancy can induce abortion.

a. Protective materials to be used for pregnancy diagnosis should fulfill LDI recommended standards

- Use plastic sleeves with criteria and standard set/specified for AI technique

- Wear protective clothing, such as overall and rubber boots.
- Use lubricant to facilitate entry of hand into the cow's rectum

b. Preparation and handling of the animal for PD

- Properly restrain before palpating the cow/heifer in a non-slippery properly constructed chute.

c. Transrectal Palpation for pregnancy diagnosis

- Perform rectal palpation with care to avoid damaging the fetus and the cow's rectum.
- Palpation can be done with either hand (Preferably use the left hand). One hand may be used to grasp the cow's tail to use as leverage to push the other hand into the rectum.
- Identify the anatomical landmarks and changes (such as asymmetric horns, ballottement) for detection of pregnancy
- As pregnancy advances, the cervix and uterus move over the ridge and into the body cavity, thus requiring the palpator to feel past the ridge and downward. Key indicators for bovine pregnancy diagnosis at different pregnancy stages is indicated in table 2 below.

3.1.6 Determining the Stages of Pregnancy

Table 2. Key indicators for bovine pregnancy diagnosis at different pregnancy stages

Months of pregnancy	Key Indicators
30-Day Pregnancy (1 month)	• Determining pregnancy as early as 30 days requires great skill and practice. However, good breeding records allow the palpator to take closer judgment by cross checking with potential physical palpable indicators. Note the following key indicators: • Uterus will be filled with fluid and feel slightly thinner. • Asymmetric uterine horns with enlargement of the gravid horn compared to the other due to presence of the growing embryo that can be felt with gentle palpation between the fingers. • At this stage, the embryonic vesicle is spherical, filled with fluid and nearly 3/4 inch in diameter • It has much the same feel as a balloon filled with water. • In most cases, ipsilateral corpus luteum can be palpated on the side of the uterine horn containing the embryonic vesicle. • The corpus luteum will be a hard teat-like projection on the surface of the ovary. • At this stage of pregnancy, the reproductive tract will still lie on the floor of the pelvis.

45-Day Pregnancy (1.5 months)	• The embryo complete attachments to the uterine wall approximately on Day 38, undergo organogenesis and develop to fetal stage. • At this stage, the fetus is nearly one inch long. • The vesicle around it is somewhat egg-shaped and measures 1 to 1.5 inches in length. • The uterine horn containing the fetus is larger and thinner walled. • The vesicle membranes fully attach themselves to the caruncles on the uterine wall. Therefore, the palpator should be careful and not move the fetus about in the uterus, but can feel the membrane slip to confirm pregnancy. Excessive movement at this time could break the attachments and cause death of the fetus.
60-Day Pregnancy (2 months)  Mouse size (ground tennis ball)	• The uterus has now enlarged until one horn is approximately the size of a banana or ground tennis ball and measures 8 to 10 inches long. • The fetus measures 2.5 inches in length • Embryonic vesicle is still prominent. • The best method of feeling the fetus at this stage is to gently tap the uterus with your hand. This causes the fetus to swing back and forth and bump against the embryonic vesicle and uterine wall that can be felt by the palpator or do the membrane slip check. • At this stage, the cervix is still in the pelvic cradle, • The CL will again be found on the ovary corresponding to the enlarged uterine horn.

90-Day Pregnancy (3 months)  Rat size	• The uterus in this stage is considerably larger because of increased fluid and fetal growth. • The fetus is now nearly 6.5 inches long and is located on the floor or brim of the body cavity. • The cervix may have pulled itself over the pelvic ridge and into the body cavity. • The stretched uterus has pulled the ovaries down. • Complete Palpation of the fetus in this stage may be difficult. • The palpatory may have to consider other factors to confirm pregnancy. Palpation of the uterine artery may be used to accomplish this. This artery, which carries blood to the developing fetus, is located in the forward fold of the ligament that supports the uterus. At this stage, it is 1/8 to 3/16 inch in diameter and has a characteristic "whirring" pulsation as the blood moves through it. It can be felt by pressing it against the left forward side of the pelvis, and can be confirmed by attempt to move it from side to side. • Care must be taken not to confuse the femoral artery, which lies in the muscle of the thigh, cannot be moved and does not have the "whirring" pulsation of the uterine artery. • Best indication at this time is the presence of the cotyledons, flattened and egg-shaped masses (measures 3/4 to 1 inch across) on the uterus, and feels slightly firmer than the uterus.
120-Day Pregnancy (4 months)  Small cat size	• The fetus measure 10 to 12 inches long and is still on the floor of the body cavity. The head of the fetus is nearly the size of a lemon and may be the first portion of the fetus that the palpator touches. • Each cotyledon is 1 to 1.5 inches in length • Uterine artery has increased in size (1/4 inch in diameter).

150-Day Pregnancy (5 months)  Large cat size	• The main change from this stage until birth is in the increased size of the fetus. • At 150 days, the fetus is the size of a large cat (approximately 16 inches long). • The uterine artery is 1/4 to 3/8 inch in diameter • Each cotyledon is 2 to 2.5 inches in diameter. • Palpation of the fetus still may be difficult because of its low position in the body cavity.
180-Day Pregnancy (6 months)  Beagle dog size	• The fetus is still deep in the body cavity. • The uterine artery is 3/8 to 1/2 inch in diameter • Cotyledons are larger. • From 180 days until birth, the fetus can be made to move by grasping its feet, legs, or nose.
Seven-Month and Longer Pregnancy	• At 210 days of age, the fetus is 24 to 38 inches long; The uterine artery with diameter of 1/2 to 3/4 inch; and the fetus can be easily felt because of its increasing size until calving

d. Other Methods of Pregnancy Diagnosis

- Ultrasound - Detection of pregnancy through the use of ultrasound may be beneficial as early as 21 days.
- Biochemical Tests - Pregnancy test can be done by different biochemical tests kits (e.g. milk progesterone test) with immediate results that can be done by producers on-farm.

3.1.7 Record keeping during Artificial Insemination

Insemination data is significant for the purpose of genetic evaluations for fertility & gestation lengths, and identifying repeating cow or holds to that serve. With main focus on these key areas, carefully maintained and accurate records that can be analyzed appropriately can determine the trends of reproductive performance, current status of

pregnancy generation, and give some guidance to where the dairy operation is heading. Therefore, every event during and after breeding should be recorded with set standards (see section 9 this guideline).

3.2 Estrus synchronization in Breeding cows/heifers

3.2.1 Animal handling facilities and Personal Protection Equipment

- Handle animals presented for estrus synchronization in properly constructed chute
- Maintain proper hygiene, septic measures and personal safety at all times, during treatment or transrectal evaluation

3.2.2 Selection of indigenous cows and heifers for synchronization

Cows/heifers required to serve as dam for crossbreeding has to be selected for synchronization and breeding should be sound based on the following points

- Physical fitness: free from physical injuries, feet and hoof problems, any disability that can interfere with pregnancy and delivery
- Apparently healthy from any concurrent infection, gynecologically sound and cycling, and without history of reproductive health problem such as reproductive tract infection, dystocia and abortion
- Parity (Age): ≤ 4 parities for synchronization and nulliparous heifers or cows with one parity for sexed semen
- Proper body condition: condition score between 2.5 and 4 (based on the scale of 1-5).
- Heifers with 60-65% of mature body weight (230 kg for crossbreed, 180 kg for local breeds)
- Milk yield performance of indigenous cow to be selected for synchronization should be at least 1kg/per day to help crossbred calves grow properly. Farmers should totally dedicate indigenous dam (mother) milk for crossbred calf.

3.2.3 Estrus Synchronization protocols

- Remember synchronization drugs are not fertility enhancers. Good results are dependent on the type of protocol, the reproductive status of the animal, and proper detection of the response. Different breeds of animals respond differently to treatments for estrus synchronization. Trans-rectal palpation of the ovary is necessary for confirmation of presence of mature CL on the ovary (Large enough to be palpated) before treatment with prostaglandin. Do not treat animals with PGF if there is no mature CL or if there is very large follicle on the ovary. Animals within one week of ovulation (Immature CL) and very close to ovulation (Static dominant follicles) do not respond to PGF treatment'
- Breeding animals can be synchronized for estrus in two ways, by bringing animals in follicular phase into artificial luteal stage (by treatment with progesterone eg CIDR implant) or by bringing animals in luteal phase into follicular stage (by treatment with prostaglandin F2α or its analogue eg. Lutalyse or Estrumate).
- Follow manufacturers prescription for optimal dose of the hormones for exotic animals, indigenous as well as cross bred animals for best responses of synchronization. The optimal doses of PGF for both indigenous and their crosses - 5ml Lutalyse (25mg dinoprost, natural PGF) given IM or 2ml of Estrumate (500 mcg cloprostenol, synthetic PGF) given IM are commonly used.
- Make a choice of the hormone treatment procedure based on the reproductive stage of the animal (Follicular or Luteal phases and non-cycling cows/heifers). Animal in luteal stage (mature palpable CL on the ovary) require treatment with prostaglandin F2α or its analogue while animal in follicular stage (palpable follicles on the ovary) require treatment with progesterone.
- Treatment, particularly with PGF, must be carried out by deep IM injection after proper disinfection of the site. Failure to do so might cause irritation of the skin or abscessation of injection site.
- A double injection of prostaglandin F2α or its analogue given 11 days apart at a dose rate 2.5ml/cow often usually gives over 90% response when treated

animals are confirmed for presence of CL in their ovaries. However, overt heat signs may not be observed on all responsive cows/heifers

- Animals responding to first treatment in a combination protocol can be inseminated up on detection of estrus.
- In procedures involving gonadotropin releasing hormones, make sure at which stage (follicular or luteal) the animal is by palpation of the ovary for CL or larger mature follicle
- The injection of GnRH (Eg Receptal) causes all dominant follicles to ovulate, recruits new follicles and luteinizes 'old' follicles. This treatment can be used in combination with PGF (given 7 Days apart)
- Use Ovsynch, Cosynch, Pre-synch, or Heat-synch synchronization protocols if a timed artificial insemination (TAI) is intended.

3.2.4 Detection of Estrus and optimal time for insemination

The following signs are common signs observed as indicators of the different stages of estrus in healthy cycling cows. Attendants/ owners should be aware of the signs of estrus to report to the inseminators. Closely monitor the manifestation of standing estrus signs for insemination.

The three observable signs of estrus are:

Pre-heat signs: restlessness, smelling and attempts to mount others, clear mucus, reduced milk production, and bellowing, red, moist and slightly swollen vulva.

Standing heat: stands still when mounted and mounting others; copious clear vaginal discharge, vulva enlarged, moist and red, nervous and excitable, and tail head roughened (the last sign could also be seen post-heat).

Post-heat (2–3 days after start of heat): moves away when mounted and clear vaginal discharge. Cows/heifers observed with these signs should be noted.

- Observe animals for signs of estrus at least twice a day (morning and late afternoon) starting 48 hrs after treatment
- For intensive/indoor production, it is better to let cows/heifers to exercise together with other cows/heifers during morning and evening so that other cows/ heifer can indicate cows in heat through mounting.
- As estrus signs vary among individuals, observation of more than two signs are always necessary for confirmation of estrus, with standing estrus being the best sign for eminent ovulation
- Consider the use of estrus detection tools such as rapid tests (eg. P4Gold, P4 Rapid test) or Vaginal electrical resistance (VER e.g., Draminiski EDC2 device) whenever available
- Use the guide in the schematic drawing to make reference of the time and best time to breed (Figure 2)

3.2.5 Time of insemination

- Animals are inseminated within 6-18hrs of standing estrus for best results of conception.
- After breeding, conception of the animal must be checked as soon as possible (3weeks for ultrasound and 60 days with manual palpation)

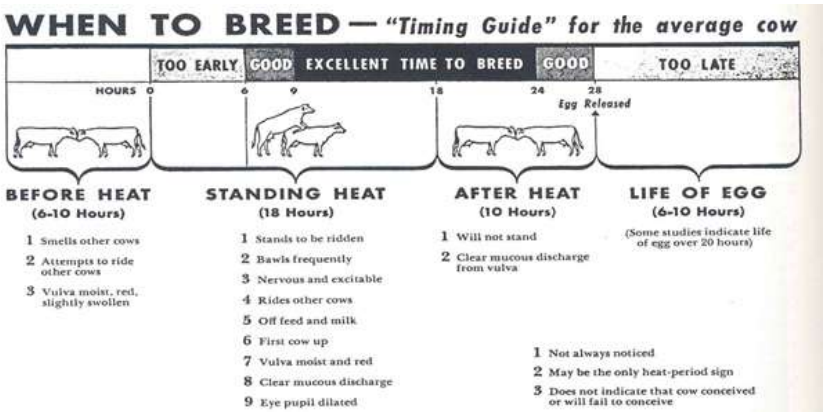


Figure 2. Decision for the best breeding time in cattle

3.3 Application of Embryo transfer technology in dairy animals

The primary use of embryo transfer is to amplify reproductive rates of valuable females. This technology is not for a wider use for genetic improvement as it is mentioned in the dairy development strategy. However, private investors who afford to import embryos and who has the capacity to implement are encouraged to apply the guideline for use of embryo transfer technology.

3.3.1 Preliminary preparation

1. **Skill and knowledge development -** The success of embryo transfer depends on factors associated with the embryo, the donor, the recipient, and the embryo transfer technician, or an interaction among these factors. Suitability of the donor and recipient is dependent on numerous management, nutritional, and estrus cycle control factors all of which demand the detail knowledge of anatomy and physiology.

NB:- Embryo Transfer is to be done only by trained expertise (in the area of Anatomy, physiology and embryology) or by veterinary obstetrician.

2. Selection and management of Donor and recipient animal

- Use properly identified and characterized breeds of cattle for specified genetic merits as a donor and recipient (Table 3)
- Do not use recipients that have failed twice in the same previous procedure (i.e., poor response to synchronization hormones).
- Both selected donors and recipients should be identified with identification Tag or a barcode to generate data and reports of individual animal.

Table 3. Selection of individual donors and recipients for embryo production and transfer

Criteria for individual animal selection	Conditions to be fulfilled
Genetic merit	• Donor Cows or heifers with better breeding values (in identified breeding program). • Cows/heifers that have lower age at first calving, high milk yield and lower calving interval should be selected (in order to index (weight) the traits 69, 14 and 17 percent can be used for milk yield, age at first calving and calving interval, respectively)
BCS	• Donor animal should be on Optimal BCS (3-4 on five scale for donor cow; 2.5 to 4 for recipients' cows/heifers) • Donors and recipients should be at good body condition and maintained on improved condition.
Gynecological examination	• Donor and recipient cows must not be pregnant; should not have anomalies, • Don't have history of reproductive disorders, (e.g., Irregular oestrus cycle, Endometritis, Cystic ovaries, etc.).
Age and Parity	• Donor cows should be between the 2nd and 4th parity; Heifers can be good recipients; Young cows seem to yield slightly more usable embryos than heifers. • Cows from the 3rd to 6th parity are suitable recipients if they have a good calving record and mothering ability; • Cows at the 1st parity are the most stressed group so are not good recipients.
Cyclicity	• Donor cows have to be greater than two months postpartum to produce a greater number of embryos;

3. Preparation of equipment and supplies for Embryo production and transfer

- Ensure preparedness of equipment, facilities and consumables that qualify standard set for embryo production, handling, transfer, preservation and/or storage.

3.3.2 Preparation of cows for embryo production and transfer

- 1. Synchronization of Donors and Recipients - The Donor and Recipient estrus cycles must be synchronized with the demonstration of "standing heat" on or about the same day.
- 2. Superovulation - The purpose of superovulation is to maximize production of oocytes thereby increasing the number of transferable embryos.
- Variability in ovarian response is related to type, dose and biological activity of hormone, duration and timing of treatments, and other hormones used in combination. Additional factors include nutritional status, reproductive history, age, breed of donors and recipients; hence choice of the super-ovulatory protocol is important. An example a superovulation protocol using swine extract Follicle stimulating hormone (FSH) is given in Fig 3.
- For Superovulation - The FSH has to be administered in twice daily doses, in reducing in volume over a period of 4 days, to obtain a total of 250 IU for Zebu, 650 IU for crossbred and 800 to 1000 IU for HF.
- Prostaglandin (PGF2∝) should also be injected on day 6 or 7 at dose rate of manufacturers prescription.
- CIDR (Controlled Internal Drug Release, progesterone source) vaginal has to be inserted for 7 days.

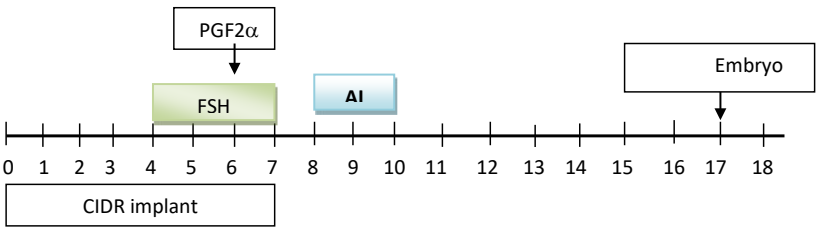


Figure 3. Donor Superovulation and Embryo recovery protocol

3.3.3 Embryo production and transfer

1. Flushing and embryo retrieval

- Properly restrain the donor cow before manipulation
- Prior to flushing response of the cow to superovulation should be assessed by counting corpora lutea on the ovaries by skilled person through rectal palpation or ultrasonography
- Prepare materials to be used for flushing (Table 4).
- Give Epidural anesthesia (2% Lignocaine, 3-5 ml) following sanitary and aseptic measures.
- Install the Flushing apparatus by introducing foley catheter with a stylet gently into the cervix. If the cervix is narrow use cervical dilator to open the cervix.
- Flush the uterus to retrieve the embryos on 7^{t}h (6th to 8th days) after AI. Flushing on the 9th day post insemination expose the embryos to physical damage as the embryos hatch and lose their protective covering (Zona Pellucida).
- Make sure that the tip of the catheter in to one of the uterine horns (Fig 4). Inflate the cuff (balloon) of foley catheter with 8 – 12 air or flushing medium to the diameter of uterine horn lumen to fully block the flow back of the flush medium, and remove the stylet.
- Flush each uterine horn 4 to 5 times; use half of the flushing fluid for each.
- Follow the same procedure for the remaining uterine horn. Care has to be made not to let out the tip of the stylet get out through the hole at the tip of the catheter and piercing the uterine wall while changing the horns.
- Cover the flushed fluid in the filter from solar radiation while taking to the laboratory
- Don't forget to inject prostaglandin to remove the CL not to maintain pregnancy development if there is missed embryos.

Table 4. Materials required for embryo retrieval

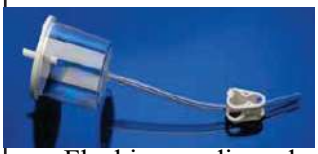
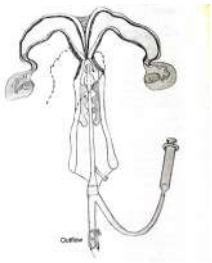
Items for in vivo embryo Flushing	**Preparation prior to flushing**
	• Make ready sterilized stylet and cervical dilator; and two 10 ml disposable syringe (one for Anesthesia, one for inflation of the cuff.
 Foley catheter	• Rinse /lubricate the folly catheter with flushing media, check the cuff for ballooning with air, insert the stylet into the folly catheter and fix with forceps,
Y-tubing 	• Fit the Y-tubing to foley catheter at the middle terminal, and to embryo filter on one end and plug the remaining terminal to the flushing medium; operate the lockers to regulate the media flow in to the uterine horn for flushing and embryo filter
  Flushing media and Embryo filter	• Flushing media is an iso-osmotic electrolyte solution, it is buffered to a physiological pH of 7.1 and is made from water purified by reverse osmosis to 18 megohms resistance; it should be free of any flakes; Warmup to the cow's body temperature (36-37^{0}C); Hung from higher level from the cows' hind part to simplify the gravitational flow of the media during flushing, and plug the terminal of the Y-tubing securely • The embryo filter has to be sterile and not damaged; Fit to the free end of the Y-tubing; check the functionality of regulating locker before starting flushing; control the filtrate flowing out;

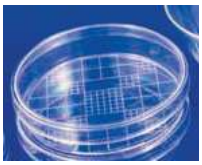
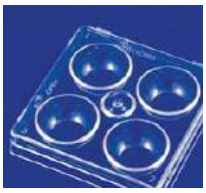
Fig 4. Level of foley catheter insertion in the uterine horn for embryo retrieval





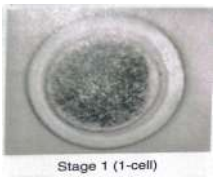
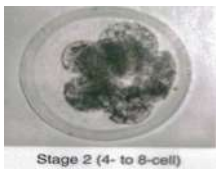
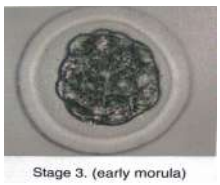
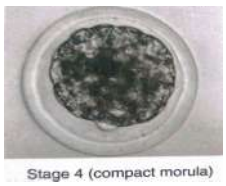
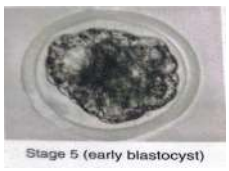
Embryo evaluation and grading is a systematic sorting of embryos determine which embryos to transfer and/or conserve, the optimal day of transfer and the appropriate number of embryos to transfer based on its form and structure with the best chance of success to develop to pregnancy. Equipment for embryo evaluation and grading and standard for evaluation and grading are indicated in table 5, 6 and 7.

Table 5. Items for embryo handling, evaluation and grading

Items	Standards of items and media for embryo evaluation and gradings
 Searching Dish	• Grided petri dish for embryo searching under microscope • Pour the flushed fluid in the filter to searching Petri dish and let the embryo to settle to the bottom of the petri dish for 30 minutes before commencing searching
 Examination	• Put the embryos in holding medium to do the embryo grading • Thoroughly examine, determine the developmental stage and make the decision to transfer, cryopreserve or discard the embryo (follow the details in Table 6 and Table 7);

Holding media 	• Holding media is support optimal embryo survival in air at room temperature; and provides essential amino acids, growth factors, enzymes, energy substrates and antibiotics In Holding Embryos can be held at room temperature for up to 9 hours or up to 20 hours at 4°C.
Freezing media 	• Freezing embryos in ethylene glycol for direct transfer; or embryos can be transferred directly into recipients after thawing, without removing the embryo from the straw, thus eliminating time-consuming rehydration steps.
Stereo microscope 	• Evaluation of bovine embryos is normally done with a stereo microscope at 50 to 100X magnification, with the embryo in a small holding dish by carefully evaluating and scoring some aspects of their appearance
 0.25 ml straw	• Label the 0.25 ml straw with the specific format that include donor ID and breed, date of flushing, development stage, quality grade. • Load the embryo in to labeled 0.25 ml straw separately and seal either with heat;
 Embryo freeze control system	• The embryo freeze control system consists of Cryochamber that stands directly in liquid nitrogen within the cryobath, and is connected to the controller which regulates the temperature of the specimens

Table 6. Stages of embryonic development with respective descriptions

Scanned image	Description of embryonic stage of development
 Stage 1 (1 cell)	Entities recovered on day 6-8 is unfertilized ova (UFO) if single cell. Make sure before discarding sometime compacted morula may mistake with UFO. Normal UFO have a perfectly spherical zona pellucida, perfectly spherical oolemma (vitelline membrane), relatively uniform granularity in the cytoplasm, and moderate volume of perivitelline space.
 Stage 2 (4-8 cells)	Any embryo recovered 6-8 days after the onset of estrus that contains 2-12 blastomeres is certainly dead or degenerate, or should not considered for transfer/cryopreservation
 Stage 3 (early morula)	Embryo at the morula stage consists of 16 cells; some of the blastomeres may be easy to visualize and some are not. This stage of embryonic development is not suited for cryopreservation.
 Stage 4 (compcat morula)	Blastomeres are tightly compacted to form a ball of cells through tight junction; the cell comprising the embryo occupy 60-70% of the space inside the zona pellucida.
 Stage 5 (early blastocyst)	Characterized by the presence of small fluid-filled cavity, blastocoele; several differentiated trophoblast cells will be visible between the blastocoele cavity and zona pellucida. The cell comprising the embryo occupy 70-80% of the perivitelline space.

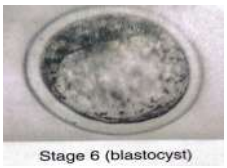
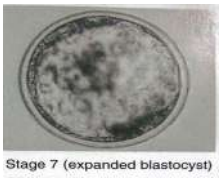
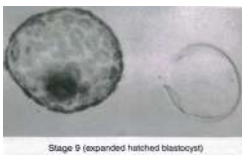
 Stage 6 (blastocyst)	Have clearly defined trophoblast layer, blastocoele cavity, and group of ICM cells; there will be little to no perivitelline space present.
 Stage 7 (expanded blastocyst)	Expanded blastocyst have increased overall diameter 12-15 times larger than its original diameter of 150-190 µm; thin zone pellucida, no perivitelline space present
Stage code 8: Hatched blastocyst	Undergoing the process of hatching or completely hatched from zona pellucida; spherical with a well-defined blastocoele or collapsed and may be difficult to identify
 Stage 9 (hatched expanded blastocyst)	This stage of embryonic development is identical in appearance to a completely hatched stage code 8 except for being significantly larger in diameter.

Table 7. Embryo quality grade assessment guide

Quality grade	Embryonic features
Grade 1 = Excellent	Grade 1 embryo is spherical, symmetrical with cells of uniform size, color and texture; or trivial imperfections such as few extruded blastomeres, irregular shape, few vesicles; At least 85% of the cellular material should be an intact, viable embryonic mass
Grade 2 = Good	Few more extruded blastomeres, vesiculation, and few degenerated cells; At least 50% of the cellular material should be an intact, viable embryonic mass

Grade 3 = Fair	Numerous extruded blastomeres, degenerated cells, cells of varying sizes, large numerous vesicles; At least 25% of the cellular material should be an intact, viable embryonic mass
Unfertilized Ovum	The nuclear mass is dark (black) and grainy in appearance; May appear round (spherical), flat or triangular in shape; Floats more quickly than an embryo (heavier) in holding medium

3. Transferring embryos

- Transfer embryos only to recipients confirmed to be on day seven post standing heat. Both the flushing of the donor and transferring to the recipient takes place 7 days after estrus
- Prior to transferring embryos check the ovaries of the recipient cattle to locate the side of the CL (left or right ovary); Transfer the embryo to the uterine horn ipsilateral (on the same side) of the ovary with the CL.
- Give due attention to prevent trauma and hemorrhage to the uterus during transfer
- Transfer fresh embryos with developmental stage 4 to 8 and with quality grade 1 to 3;
- Load to the transfer gun, cut the sealed end, cover the sheath, and deposit in to the uterine horn with the active CL following the standard sanitary procedure. The embryo can be deposited in the middle third of the uterine horn.

4. Freezing of Embryos

- If there are no recipients, proceed with freezing the embryos for later use;
- Make necessary adjustment and preparation of embryo freezer before transferring embryos to be frozen from holding media to freezing media;
- Transfer embryos at developmental stage 4 to 8, and with quality grade 1 and 2 from holding media in to freezing media (Ethylene glycol) and immediately load in to labeled 0.25 ml straw and immerse in to embryo freezer with liquid nitrogen (This step varies with the model of embryo freezer and follow the manufacturers instruction for embryo freezing)

- Subsequently, put the loaded straws into labeled goblet or canister with liquid nitrogen (similar to that for frozen semen) for long term storage.

4. Reproductive Health management and Biosecurity measures

Implement evaluation of the key performance indicators (KPI) and target values/ reproductive goals used to measure performance. Each producer should establish reproductive goals for the herd (age at first breeding of 13 – 15m; age at first calving of 24m; Days open of 85 Days; First service conception rate >50%; Services per conception of < 1.7; incidence of Abortions - < 4%, Metritis/Endometritis < 10%, Cystic ovaries < 10%, Repeat breeders - < 15%, Retentions of fetal membrane < 8%, still birth 5%, Calf mortality <10%),

4.1 Reproductive health management of breeding animals

Infectious and noninfectious diseases of reproductive system can be minimized by sound health management and skillful intervention.

- As a first step of intervention enhance knowledge and skill-based reproductive health management and intervention through awareness creation, training on application of modern technological tools and procedures
- Upgrading producers and service providers understanding of epidemiology and risk factors for major diseases affecting reproductive health and performance
- Hands on standard biosecurity practices, information exchange among producers and services givers
- Upgrading diseases control and prevention capacity through availing / accessing drugs, diagnostic kits and other inputs for the sound reproductive health management and optimal performance
- Developing strategic diseases control and prevention scheme, centralized information and data exchange mechanisms and platforms

4.1.1 Regular Reproductive health Inspection and Corrective measures

- Routine postpartum examination – All fresh delivered cows have to be checked and rechecked within 15 to 45 days; Cows with endometritis should be treated with antibiotics, Ecbolic drugs such as oxytocin, PGF2α and supportive treatments. Systematic use of PGF2α at 5 and 7 weeks postpartum may improve pregnancy rate. The economic benefit of efforts to identify and treat endometritis is herd-specific
- Pregnancy examination – all cows with AI service record has to be checked for pregnancy within 60 days;
- Cows with abnormal cases of reproductive problems such as abortion, RFM, abnormal discharge that persists > 2 weeks, postpartum cow over 50 days with no estrus or abnormal length of estrous cycle, repeat breeders or animals serviced $\geq$ 3 times & not pregnant have to be checked with routine herd inspection
- Monitor incidence of mastitis and apply routine control practices (see annex for detail procedure)
- Monitor cases of metabolic disorders and nutritional problems, and apply control interventions
- Monitor calf/heifer health problems and mortalities
- Control and prevention of lameness: Conduct early lactation claw trimming or regular hoof trimming at least once in 6 months; Check hoofs regularly for any injury and signs of laminitis; Adequate lying space should be provided to avoid excessive standing; Increased routine footbath use, avoid using of footbath with an astringent and antiseptic solution such as copper sulphate 5% and zinc sulphate 10%; Use of flooring that reduces or prevents slurry accumulation in yards; Provide systemic therapy,
- Set regular vaccine programs for common infectious diseases and strategic parasite control

4.1.2 Prevention and case management of repeat breeder

- Ensure that cows are served at the correct time; Use proper AI technique and procedures as described in this guideline; Don't inseminate cows with abnormal discharge.
- Don't inseminated cows at the first postpartum estrus
- Minimize stress at AI service. For example, try and avoid serving around turnout or when you change the diet.

4.1.3 Biosecurity measures

Bio-security refers to procedures and practices that reduce the transmission of pathogens from outside sources into the farm through visitors, feed deliveries, harvesters, and equipment or pathogens between areas of the farms.

- Use standard set of personal protective equipment (PPE), clothes and boots for routine services
- Conduct periodic clean up and disinfection of houses to break the life cycle of disease and its transmission; Avoid accumulation of wastes and manures that serve as a source of infection.
- Control pest and rodent population to reduce the incidence of disease transmission such as lumpy skin disease. Insecticide is used in this case.
- Use well-ventilated housing to reduce suffocations and susceptibility to respiratory distress (pneumonia)
- Avoid mixing of animals with different age groups, watering and feeding areas in order to minimize overcrowding and bridging of pathogens.
- Install disinfection pit at the farm gate to disinfect vehicles and visitors' shoe.
- Periodic clean out, clean up and disinfection of houses and equipment.
- Ensuring availability of detergents, disinfectants and antiseptics and cleaning equipment in the farm.
- Movement restriction between and within the farms to reduce pathogen

spread

- Apply proper hygienic and biosecurity measures in the cow's environment and feed storage. Feed which is contaminated with molds should not be fed to cows; keep vermin out as they can spread bacteria and viruses, particularly Salmonella.
- Minimize the risk of introducing diseases onto the farm. Quarantine newly incoming cattle for a minimum of 14 days. Isolate sick animals and properly dispose hazardous wastes, manures and dead body and associated contaminants.
- Pay special attention to the health status of breeding bulls to stop the spread of venereal disease.
- Ensure farm visitors and their equipment are clean before they enter the farm. Install disinfection pits.
- Make sure your boundaries are stockproof.
- Isolate aborting cows and immediately remove of aborted materials to reduce spread infectious agent within the herd.

4.1.4 Management of vaginal and uterine prolapse in breeding cows

Uterine prolapses occur after calving when the uterus inverts itself through the cervix and the vagina out of the cow. Various predisposing factors have been suggested for uterine prolapse in the cow, i.e., hypocalcaemia, prolonged dystocia, fetal traction, fetal oversize, retained fetal membranes, chronic disease and paresis. Older animals are more commonly affected. The prognosis is generally favorable if the cow survives having the uterus replaced (See annex 4 for the procedures and steps to re-place the prolapsed uterus).

4.2 Management of hypocalcemia in breeding cows

- From 21 days prepartum, reducing dietary cation–anion difference (DCAD) (starting from 7.4 mmol/100 g to 16 mmol/100 g of dry matter) is shown to avert the risk of hypocalcemia and associated reproductive problems.
- Clinical milk fever cases should be treated with IV calcium gluconate (23%, 500 mL = 10.8 g of calcium).
- Use other supportive medications such as multivitamins and fluid therapy (Use glucose 50% solution; 500ml IV followed by 20% solution subcutaneous)
- Minimize calcium supplements in the feed of animals during late pregnancy; use propylene glycol 125-250gm mixed with water

4.3 Management of mastitis in breeding cows

Mammary gland infection causes immune responses, resulting in the abnormal secretion of cytokines and hormones and abnormal function of the reproductive system such as the ovary, corpus luteum, uterus and embryo. Thus, cows with mastitis can have delayed oestrus, decreased pregnancy rate and increased risk of abortion. Therefore, cases of mastitis have to be regularly checked and monitored by screening technique and tests such as strip cup detection (Clinical mastitis), California Mastitis Test (CMT) or Soda test (see annex 3).

4.4 Common infectious and parasitic diseases

Common infectious diseases that affect health and animal welfare situation (e.g., febrile cases) can directly or indirectly affect reproductive performance of dairy animal. Active and passive immunization is integral part of sound herd health management that should be routinely practiced. Passive immunization (e.g., in neonate calves) produces temporary immunity by transferring antibodies from a from the dam to the offspring via colostrum feeding to gives protection during preweaning age. Therefore, to minimize

the effect of some contagious and devastating diseases require prior planning and instigation of control and prophylactic measures (Table 8). Parasites are the most important health problems that significant cause decrease in productivity and young stock mortality. And hence, strategic mass treatment (Table 9) has to be considered in addition to individual animal treatment as part of health program. Besides, it is imperative to undertake regular testing for some economically important diseases of dairy cattle for better planning and herd health management (Table 10).

Table 8: Common types of vaccines and calendar of vaccination

No	Disease type	Brief description	Treatment and prevention
1	Anthrax	Highly infectious and fatal disease-causing sudden death	• Administer Penicillin at a rate of 44000IU/kg Or use other available antibiotics • Vaccinate once a year
2	CBPP	Acute and subacute diseases of cattle caused by Mycoplasma mycoides subsp.mycoides	• Treat with Tylosin (10 mg/kg, IM, bid, for 3 days) or danofloxacin 2.5% (2.5 mg/kg/day for 3 days) • Vaccinate once a year
3	Bovine Pasteurellosis	Acute fatal disease of cattle caused Mannheimia haemolytica and Pasteurella multocida	• Early treatment with antibiotics and flunixin meglumine is helpful • Vaccinate twice a year
4	Black leg vaccine	Highly fatal disease of young cattle resulting blood poisoning leads to rapid death	administration of penicillin (10,000 IU/kg, IM) Vaccinate once a year

5	Lumpy skin disease vaccine	LSD is highly contagious viral disease of cattle with inapparent to severe occurrence	• Has no direct antiviral treatment; Use of antibiotics, painkillers, and wound care sprays to treat symptoms • Vaccinate once a year
6	Foot and Mouth disease vaccine	FMD is contagious diseases, and is not normally fatal to adult, but can be fatal to young calves;	• Has no direct antiviral treatment; Use of antibiotics, painkillers, and wound care sprays to treat symptoms • Vaccinate once a year
7	Bovine viral diarrhea (BVD)	Acute viral diseases of cattle with high morbidity and mortality	• Has no direct antiviral treatment; Use of antibiotics, painkillers, and wound care sprays to treat symptoms • Vaccinate once a year
8	Infectious Bovine Rhinotracheitis (IBR)	IBR is a highly contagious viral disease that affects cattle of all ages	• Has no direct antiviral treatment, antibiotic therapy designed to control secondary bacterial infection • Vaccinate once a year (if vaccine is available)

Table 9: Implementation of deworming and spraying strategies in the central highland

Category	Description	Treatment and control
Endoparasites	The main GIT nematodes, trematodes, flukes and protozoa's (Coccidia) are interna parasites of economic importance in cattle	GIT nematodes (Trichostrongylus, Strongyloides, and Oesophagostomum), flukes (Fasciola's), Coccidia, and other internal parasites, Antinematodal drugs (benzimidazoles – albendazole, fenbendazole, oxfendazole), macrocyclic lactones (Ivermectin), imidazothiazoles (levamisole); Sulfadimethoxine (50 mg/kg, as a drench or in water); Sulfamethazine (247.5 mg/kg, in the drinking water or as a drench),
Ectoparasites	The main ectoparasites encountered in cattle are flies, lice, scabies, and ticks	Mites and lice are controlled by washing the infected area, spraying or dipping the animal with a suitable treatment Synthetic pyrethroids (Fenvalerate, Deltamethrin 10%), Diazinon (200g/L) – dipping or spray, Ivermectin 1% (against internal worms – 1ml/50kg SC),

Table 10: Regular testing of animals for early disease diagnosis and screening

S/No	Types of Diseases	diagnosis and screening techniques	Remark
1	Brucellosis	Rose Bengal plate test (RBPT), Complement fixation test (CFT)	Segregate and cull reactors
2	Bovine TB	Intradermal tuberculin test (CIDT)	
3	Mastitis	CMT and White side test	
4	BVD and Bovine herpes virus V1	C-ELISA	

4.5 Calves Health management

Maximizing performance, health, welfare and profitability of replacement heifers can be achieved by focusing on the first 60 days of the calf's life on feeding and housing management. The most common health problem includes acute diarrhea (bacterial and viral diarrhea), omphalitis, pneumonia and arthritis. And hence, designing control strategy and implementation is imperative (see under Care of the newborn section).

4.6 Animal Health Records

Recent disease epidemics have drawn attention on livestock disease and their potential to harm development. This is because they reduce herds and flocks dramatically; cause trading partners to quite understandably and put trade barriers; they are a deterrent to sustained livestock production; and add significantly to the cost of livestock production through costly disease control measures. Reproductive diseases are those that have a primary or significant negative impact on fertility including Venereal diseases transmitted through coitus or other infectious diseases such as TB and mastitis. Therefore, capturing data on farm in real-time helps to identify where the causes of infertility lie. The routine record keeping should consider Disease occurrence and date, all measures taken as curative and prophylactic treatments, Vaccination, Dipping/spraying for exter-

nal parasite, mass deworming, and Post-mortem examination and findings. Recording formats indicated in the Annex 2.

5. Breeding of dairy cattle

Breeding value-based selection is the best method for genetic improvement of animals in dairy production. Selection criteria for individual animal to be used as a breeding dam is indicated under the guidelines set for estrus synchronization and AI, donor and recipient animal selection (Section 3 in the document).

5.1. Indigenous cattle breed improvement for Mixed crop-livestock production system

An open nucleus breeding scheme that has two tiers (breeding and production unit) is suggested as an option for genetic improvement of indigenous breeds.

- Existing ranches or new nucleus herds can be used as breeding unit (tier 1) (Fig 5).
- Community cattle should be involved in the program as a production unit (tier 2).

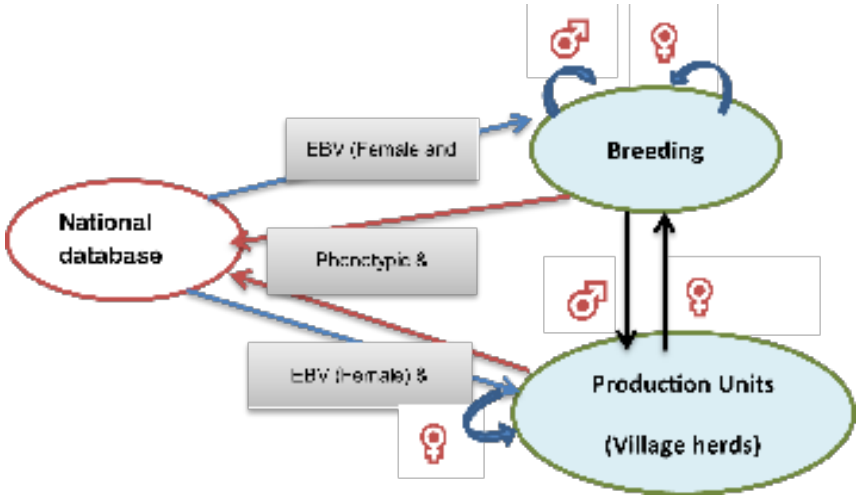


Figure 5. Schematic explanation for two tiers using indigenous cattle genetic improvement scheme (EBV= estimated breeding value; genetic contribution of the selected cows/heifers that can transfer/add to next generation estimated based on own and relative information)

5.2 Indigenous cattle breed improvement for Pastoral production system

In pastoral area dissemination of improved germplasm can be undertaken through estrus synchronization and artificial insemination (as indicated under 3.2 in the document).

- Training should be provided to pastoralist on genetic improvement, reproductive biotechnology and associated husbandry practices.
- Participant pastoralist should select 5 to 7 best cows/heifer from his herd and provide special management for three months prior to selection by dairy experts. Out of these, three to four cows/heifers should be chosen by experts based on criteria set to select cows/heifers for synchronization and AI (under section 3.2).
- The selected cows/heifers can be brought to ranches and receive better management for one to three months. Synchronize these cows/heifers to induce heat and breed by selected indigenous bull or inseminated with semen collected from superior bulls selected for breeding program within 3 months.
- Some cows/heifers that come to heat under natural circumstance can also be inseminated.
- Finally check for pregnancy 60 days after insemination. Those cows/heifers failed to conceive and become pregnant within given period must be delivered back to owners.
- Pastoralist should be advised to grow male calves born from the selected and pregnant cows to use as a bull to improve the rest of his and neighboring herd and cull cows/heifers failed to conceive.
- Avoid mating of genetically closely related animals to reduce inbreeding

5.3 Urban and peri-urban dairy

1. Commercial dairy production system

There are very few large-scale dairy` farms that are distributed in urban and peri

urban areas. Some of these farms are specialized in milk production by keeping mainly pure exotic dairy breed.

- Straight breeding with open nucleus scheme is suggested for commercial dairy production system.
- The details of breeding program should be prepared according to dairy breeding strategy and breeding program guideline.

2. Small holder dairy production

Most of the small holder farmers are engaged in milk production by keeping mainly crossbred dairy cattle. Friesian and Jersey sires' semen can be used from the sire side in areas where crossbreeding has to be conducted. Jersey crossbreeding is recommended in area comparatively where feed is less available and low access to fluid milk market.

A. Cross breeding for Medium to high input-output milk shed

- Cross breeding would be the potential alternative breeding option the improvement
- Mating plan as suggested in Fig 18 helps to improve crossbred population and develop new (synthetic) breed in long term for medium to high input-output milk shed areas such as Holetta, Selale, Debre Brehan and Asella peri-urban and rural areas.
- Different crossbred herds found on the hands of research centers, higher learning institutes and nuclease herds on different ranches (new ranches to be established if necessary), can be used as breeding unit (tier 1).
- Indigenous cows should be inseminated with exotic semen to produce first generation, then follow the mating plan shown in Fig 6.
- Any crossbred dairy cattle kept by smallholder farmers in peri-urban and rural area has to be considered as production unit (tier 2) (Fig 6) and should be mated with 75% exotic inheritance sires (semen) selected from nucleus herd (tier 1).

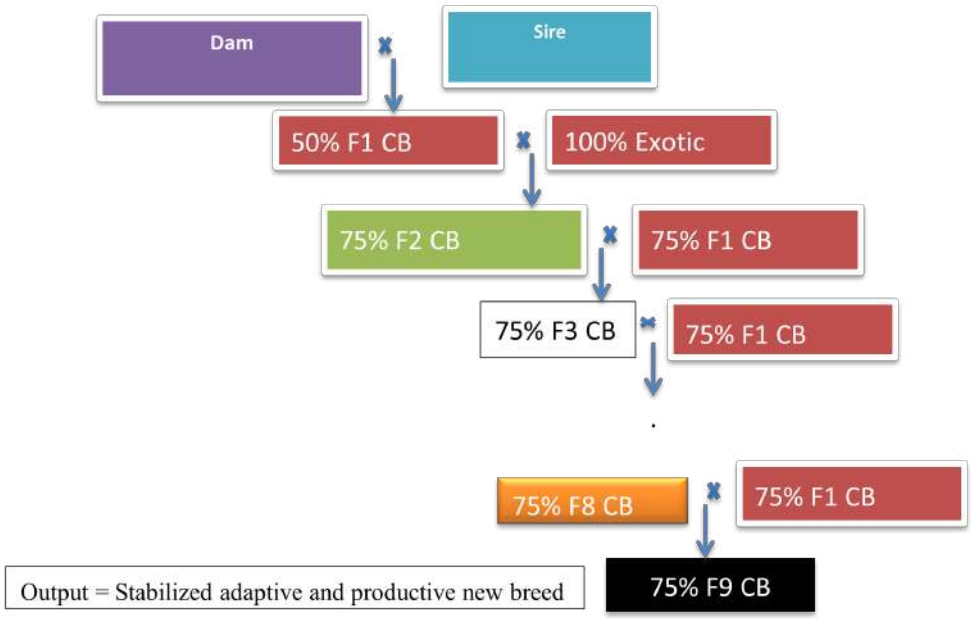


Figure 6. Mating plan suggested for medium to high input-output milk shed areas delineated for crossbreeding

B. Cross breeding for Low input-output milk shed

- Fifty percent exotic inheritance crossbred cattle are more suitable for low input - output areas.
- The next generation should be maintained at the same breed level using selected 50% F1 bulls (semen from F1 bulls) for all crossbred cows and heifer in order to develop composite new breed in long term (Figure 7).
- Indigenous cows should be inseminated with exotic semen to produce first generation. Starting from F1 crossbred cows inter-see mating (straight breeding) procedure to be followed. All generation of crossbred cows and heifers should be mated with sires (semen) with 50% exotic inheritances selected from tier 1.

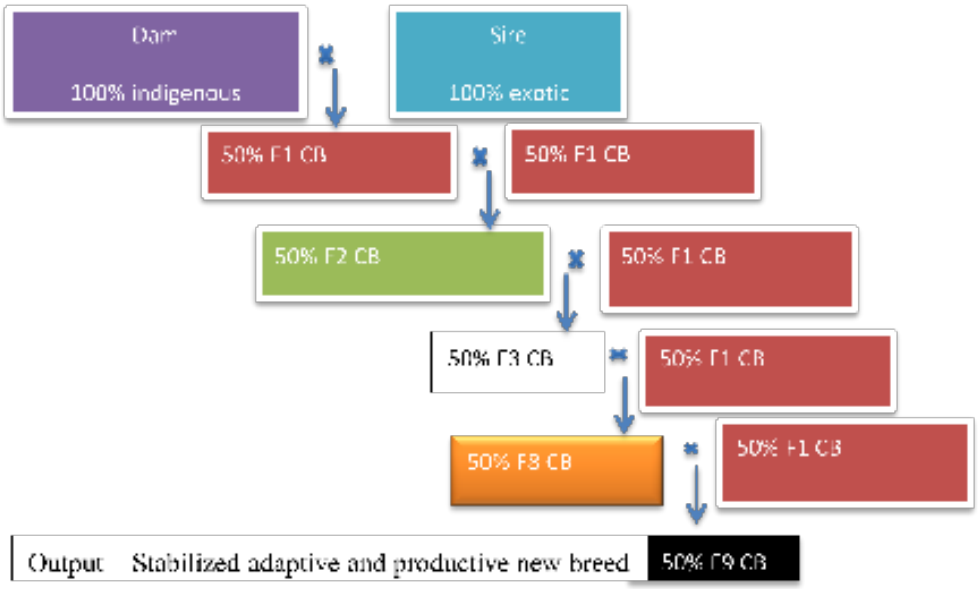


Figure 7. Mating plan suggested for low input-output milk shed areas delineated for cross breeding

C. Other breed improvement options

In absence of breeding program, the following points can be taken as breeding option to reduce negative selection and bring genetic progress at farmers animal level.

- Select best performing cows (provide better milk yield, grew fast and have lower calving interval) as a dam for bull calve production or use bulls born from better cows
- Select better performing cows (in terms of milk, growth and reproductive performance) as dam for heifers' replacement
- Bulls with retarded growth should not be used for breeding or should be culled from the herd
- Cull poor performing cows and calves with slow growth rate.
- Keeping calves and heifers born from better cows for herd replacement
- Avoid mating of closely related animals to reduce inbreeding

5.4 Record keeping

Keeping records refers to collecting relevant information that can help farmers to make good decisions and to keep track of activities, production, and important events on a farm. For the detail of record keeping see Annex 2.

6. Husbandry of breeding animals

Sound breeding animal husbandry aims at improving welfare of breeding animal for optimal reproductive performance and lifetime productivity, yielding more lactations and calves by each breeding cow. Husbandry in the context of this guideline include correct feed resource, feeding management, and proper housing

- The types of feeds commonly used for breeding animals include roughages, concentrates, vitamins, minerals and water
- Stored feed must be always kept dry and protected from moisture and fire; hence it needs to be stored in a well-ventilated area to avoid mold development and exposure from direct sun light. Consider storage of feed during the time of surplus to avoid seasonal variation/fluctuation in feed availability and quality.
- Mixing of feeds should be carried out in quantities that is sufficient for at least a week or better for one month at most.

6.1 Types of animal feed resources

- **Roughage feeds:** They are usually fed ad libtum with 10% allowance for refusal. Roughages include grasses from improved forages, pasture grasses, hays, silage, root crops, straw, and stover, etc.
- **Concentrate feeds:** They are fed separately as an ingredient or a mix commonly balanced for energy and protein to meet nutrient requirements that are not met by roughage diets. Concentrates of high energy value (>12 MJ, ME/kg DM) include energy-rich grain and grain by-products (maize, sorghum, wheat, barley, teff, oat and wheat bran, rice bran, maize gluten, etc.) and protein-rich

(>18 % CP) (oil seed cakes such as soybean, cottonseed, peanut, Noug seed cake etc.) and by-products from the processing of sugar beets, sugarcane, animals, and fish meal) feed sources. Excess daily allowances could result in metabolic disorders, in addition, to economic losses.

- **Minerals and vitamins:** The most commonly used mineral under local condition is salt. It is often included at a rate varying from 1 to 3% in dairy cattle's ration. In addition to salt, Di-calcium phosphate mineral is usually recommended to supplement growing calves, high producing, and pregnant cattle. Leguminous forages, molasses and some local water and soil types (Bole, Megado etc.) are also very good sources of minerals. Vitamin requirement can be met through regular feeding of 1-2 kg green forages. However, in extreme cases, pre-mixes containing both minerals and vitamins can be used to overcome deficiencies and thus improve reproduction and productivity in dairy cattle.
- **Drinking water:** Dairy cattle must have ad libtum access to clean and sufficient drinking water at all times.

6.2 Feed and feeding management of breeding dairy cattle (Feeding regime for different age group indicated under annex 4)

Providing balanced nutrition is the foundation for productive dairy animals and profitable dairy business; without which animals cannot express their full genetic potential nor be efficient in performance. The types of feed resources, their proper storage, standard mixing and implementation of standardized feeding methods for different breeding cattle are components of the feeding management

- Roughages, water and mineral block leak shall be given ad libtum
- Dairy formula feeds (formulated concentrate mix) are offered at the rate of 0.5 kg per each liter of milk production during early lactation (72 to 80 days) and 0.3 kg per each liter of milk production during mid and late lactations (80 to 305 days).
- Flat rates such as 0.5 per kg per milk production is often recommended across

the entire lactation periods (early to late lactation) for cows maintained on very poor quality basal diets including crop residues.

6.3. Feeding management of heifers

- Mature heifers include the age between one year to first calving
- Replacement heifer diets should be formulated with adequate Metabolizable energy and protein to meet the demand for lean growth. Excess energy intake should be avoided pre-pubertal for replacement heifer to reduce adipose tissue deposition.
- The heifers feed options should be based on a concentrate-to-roughage ratio of 50:50 that can be used to support the optimum growth of cross-bred dairy heifers. When feeding heifers, the farmer should aim to achieve a growth rate of 0.5-0.7 kg/d and first calving at 22 to 24 months of age. Mature heifers reach target live weights (230 kg) for breeding at 14-16 months of age.
- If allowed to graze good quality pasture alone, it can fulfill the nutrient requirements of heifers (growth and maintenance). In the absence this supplementation with concentrate should be made at 1% of their body weight.
- Drinking water and mineral salt supplement is recommended on a free-choice basis.
- Pregnant heifers may be maintained on good quality forage alone but concentrates should be given if the forage is of low quality.

6.4 Feeding management of calves and immature heifers

- Calves should be offered only high-quality forages early in life and supplemented with concentrates (calf starter)
- Calves and immature heifers should be offered fresh water ad libitum in addition to milk.
- Liquid milk or milk replacer is reduced from 3 weeks of age to encourage the calf to consume and maximize intake of dry feeds.

- Calves must be provided with adequate clean drinking water at all times
- For attaining physical maturity, growing calves and immature heifers must be fed according to feeding standards annexed in this guideline.

6.5 Feeding silage-based diets to dairy cattle

The quality of the silage depends on the variety, harvesting stages and the ensiling process. Similarly, the amount of concentrate to be fed depends on animal species and physiological conditions of the animals, and silage quality.

- With good quality silage offered as a sole roughage or TMR diet (the silage usually made from whole maize or sorghum crops) the daily allowance of concentrate for early lactating cows would be 1kg concentrate for every 3 lt of milk produced by the cow.
- There is no need to supplement dairy cows and mature heifers on such quality silage with concentrate. If quality of the silage is compromised, even mature heifers and dairy cows meant for breeding should be additionally supplemented with good quality concentrate of 18-20% CP at the rate of 0.7 to 1% of their live weight.

6.6. Dairy animal housing and housing requirements

Dairy animal housing needs meticulous selection of the site, proper infrastructure and house construction. The dairy housing structure has to be comfortable enough for various age groups, physiological status, breeding, calving, isolating sick cows and quarantine of incoming cows in order to efficiently produce and reproduce. Besides, sanitation is necessary in the dairy housing in order to eliminate all microorganisms that are capable of causing disease. Therefore:

- The type of housing is a matter of choice where it is affordable, dairy animals should be kept in a conventional dairy house. Alternatively loose house can also be used particularly when there is sufficient space and warmer area (Figure 8 and 9)

- The housing should have proper facilities and components like milking parlor, maternity/calving pen and calf pens, isolation and quarantine rooms, stores, feeding-box/manger, drinking area and loafing area with easy access to potable water and resting places, and provided with enough light and proper ventilation.
- The housing should be suitable for feeding, heat detection, animal handling, for sanitation and maintenance of hygienic conditions.
- Stalls should provide comfortable and adequate resting surface for cows to lay at least 8 hours per day.
- Floor system in conventional housing should always be non-slippery, and preferably with mat or bedding
- Watering and feeding trough must be constructed in such a way that it prevents stamp on/trampling by feeding animals
- Detailed specification of dairy housing construction is found in the annex in this guideline

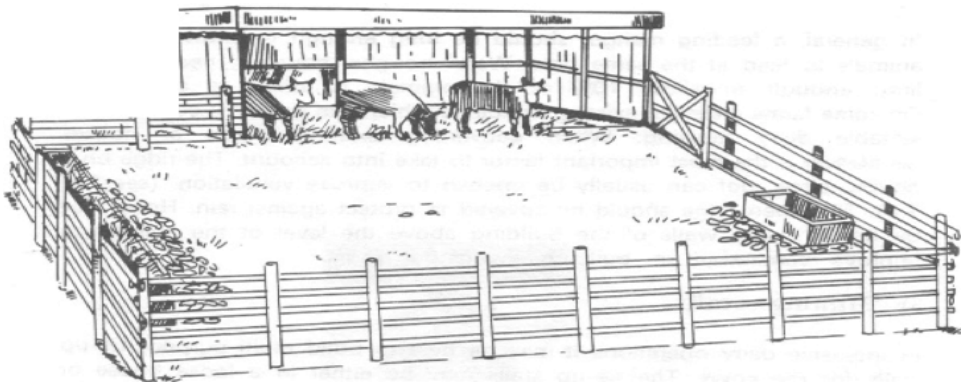


Figure 8. typical loose house (source: Felleke, 2007)

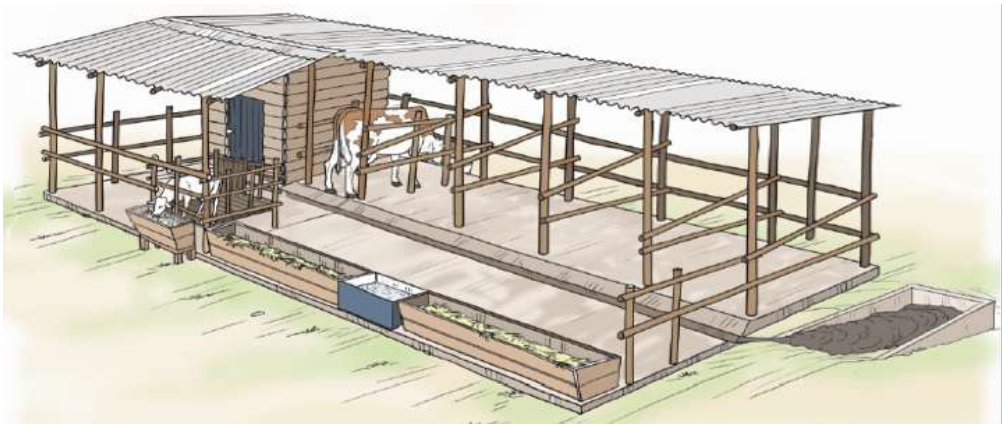


Figure 9. A zero grazing/conventional barn complete with a sun shade structure

6.7 Calf management

The management for healthy calves starts at the pre-calving period. Neonatal care begins with timely feeding of adequate amount of good quality colostrum to ensure the passive transfer of immunity. Besides, to ensure the welfare of neonates, the calving environment should be well bedded, clean and hygienic, and well-ventilated to minimize risk of occurrence of diarrhea, pneumonia and other neonatal diseases.

6.7.1 Care of the newborn

- Immediately after calving, clean the nose by squeezing with clean rags or hung down to let the fluid drain from the nose, and let the calf to the dam for licking and cleaning. If the cow can't lick, clean the calf with rags and keep the newborn calf warm
- Dip the navel with tincture iodine to disinfect immediately after birth.
- Make sure that the calf is properly identified and record the condition of delivery and other related important data

6.7.2 Colostrum feeding (1-4 days)

Colostrum is vital in providing passive immunity to the newborn calf. It also contains a higher amount of important nutritious diets and vitamins which are essential to combat disease.

- Allow the calf to suckle or feed colostrum to the newborn calf within 30 minutes to one hour after birth. Colostrum is critical to the survival of the newborn calf, and attention should be given to the time, quantity, and quality of colostrum.
- Calves should receive 5 to 6% of their body weight (1.5 to 1.8 liters for a 30-kilo calf) as colostrum within the first 6 hours and again within the subsequent 6 hours (See Annex for more detailed provision of colostrum).
- If the cow's production of colostrum is insufficient, colostrum from other cows or replacer colostrum supplements may be necessary.

6.7.3 Housing the newborn calf

- Calves should be kept in individual boxes during the first three weeks of age. The dimensions of the box should be 1.5 m length and 0.8 m width. Calves can be transferred to group pen as they grow older (Fig 10).
- Calves should be kept confined to these pens until weaning and thereafter allowed outside during the day.
- Calf pens must be solid, non-slippery/well-drained with clean bedding (straw, coffee husk, rice hulls, or wood chips) warm and dry, and easy to clean and provide secure footing, and the floor allows the claves to stand, walk, lie down, and pass excreta.
- Cleaning and disinfection of calf pen should be carried out at the beginning and end of each calving season, and during periods of disease outbreak
- Dusty straw materials or clamped with fungal growth and others should never be used as bedding material for calves

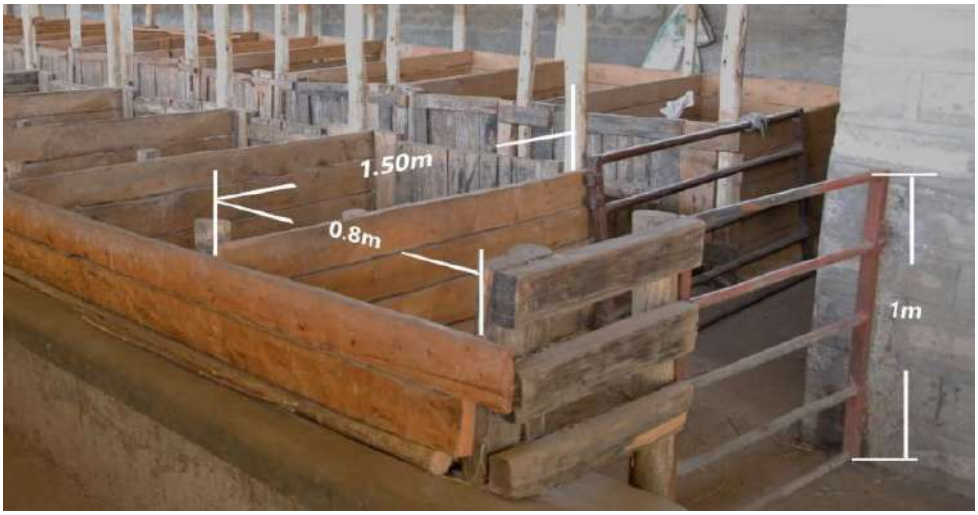


Fig 10. Individual calf pen for neonatal calves

6.7.4 Health care for calves

Newborn calves are susceptible to many infectious agents since their immunity is not well developed. Even with provision of colostrum, there are a number of factors that predispose calves to various diseases such as diarrhea and pneumonia which are the leading causes of mortality.

- Maintain a good hygiene of the calf pen at all times to prevent diarrhea and pneumonia
- Isolate sick animals and install treatment as soon as possible
- Provision of parenteral antibiotics and fluid replacers will give good results if they are applied timely
- Start to vaccinate calves after 4 months of age and apply regular deworming once the calves start grazing

7. Dairy Farm Recording

Good animal records are a vital part of animal management, allowing for evidence-based decision making and records provide an ever-expanding source of data and

information upon which to base breeding, health and husbandry. It also good key performance indicators without which decision making would be difficult particularly tracking animal & evaluating performance. Standardized recorded data provide up to date information on herd dynamics and performance over a year, identifying problems and gaps, and setting future goals/directions, and hence, record keeping at different level would be compulsory. On the contrary, records should not be more troublemakers than they worth, and can be done by using cards in Files, wall charts (especially for reproduction) and a computer. The record should satisfy the following criteria:

- **Records must be useful** – should be prepared and kept in a simple and systematic form that they can be easily converted into information and lead to actions.
- **Duplication must be avoided as much as possible and formalize**. An individual farmer should follow and maintain recording format recommended by extension officer.

7.1. Types of dairy farm recordings

There are different types of dairy farm recordings. The following list provides the most applicable types of dairy farm recordings at smallholder farmer level in the Ethiopian context:

- **Animal Identification/history record** - provides information about major events occurred on individual dairy cattle from birth up to the time it leaves the herd. It includes Animal Name or ID, date of birth/purchase, breed type, calving period (month) for each calving interval, lactation period (days) for each lactation, lactation yield (average per day and total yield), date of drying, calf gender and ID number, age when culled, date of disposal (sales/death).



- **AI service Record** – help as a guidance to determine the success of AI service and profitability of the cow calf operation when accomplished with accuracy and consistency. It shall include owner's profile, Animal identification number/ name, Dates heat observed; Service Dates; Sires ID and Breed; Technician`s ID; Service number (First or Repeat); pregnancy examination, previous date of calving; expected calving date, drying off date, and any additional remarks (Table 2).



- **Breeding Records –** it should include ownership detail profiles; Name and/or Identification number of animals; Identification method (Ear Tag, tattoo etc.); Birth date or date purchased and Breed Type. Pedigree Information such as extended family (Names & ID), Dam; Sire; Sire`s Sire; Sire`s dam; Dam`s Sire and Dam`s Dam, and Breeding parameters such as Age at First Calving, Parity, Date of Calving; Days after last calving, Calving complications; Calving Interval, reproductive diseases/disorders (if any) have to be recorded.

Table 11. AI data recording sheet

Individual Cow AI Record

IAEA/AFRA Project on Improving Milk and Meat Production

Farmer: _____ Farm: _____

_____ Address: _____

_____ _____

Cow ID: _____ Breed: _____

Breed of Sire: _____ Breed of Dam: _____ Milking only: Yes/No _____

Birth Date: _____ Lactation No: _____ Suckling: for letdown only once per _____ day/twice/day

Last calving date: _____ Remarks: _____

AI No	Date	Heat to AI (hr)	AI time (am, pm)	Service of AI (U/C/V*)	Bull & Breed	Semen Batch	Date of Milk sample			Result of Milk Progesterone (mmol/l)			PD date & Result	Remark
							1	2	3	1	2	3		

(*U = Uterus; C = Cervix; V = Vagina)

Name of Inseminator: _____ Signature: _____ Date: _____

Laboratory Interpretation and Recommendation:

Name: _____ Signature: _____ Date: _____

- **Health (Disease and treatments) Records** - Information management is fast becoming the key to effective action in animal care and keep track of the disease events in each animals' lifetime; guide to better management practices by leading the attention to repeated events or seasonality or certain vulnerable groups of animals over time (e.g., it can show how animals almost always need disease treatments in different seasons). Accordingly, health record should include disease occurrences, treatments and control measures practiced, events that affect welfare of the cows that interferes with reproductive performances and calves' survival (Table 12, 13, 14).

Table 12: Animal health observation and treatment record

No.	ID/ Name	Date of observation	Major signs observed	Suspected disease	Treatment given			Response	Remarks
					Treatment	Duration	Dosage		

Table 13: Vaccination record

No.	ID/ Name	Date of vaccination	Type of vaccine used	Remarks

Table 14. Mortality record

No.	ID/ Name.	Age (months)	Sex	Date animal died	Value of animal	Remarks

-

- **Production Record -** Milk production record captures the individual cow milk yield per day, and has to include milking cow name or ID number, daily milk yield, and start & end lactation period/date.





- **Feed & Feeding Record** – dictates about feed produced and/or acquired, storage and treatments, and feeding system. Feeding records give information about the amount, type and quality of the feed provided to dairy animals, type and quantity of on farm feed produced and feed purchased from outside sources, and events happened following feed changes.

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Annex 1

Procedures and steps in dairy cattle AI

A. Thawing and loading the semen straw

- Insemination supplies should be kept dry and clean at all times. Breeding sheaths should be stored in the original package until used.
- Open the container and plunge the tweezers (forceps) into the liquid nitrogen for cooling.
- Bring the canister containing the straws carefully to the neck of the container and remove the straw with the cooled forceps.
- Remove the straw with tweezer and thaw at 37^0C.
- Place the straw into the thermos flask containing warm water (37°C) as quickly as possible (<3 seconds) and thaw for 30 seconds.
- Remove the straw from the water and dry it properly using paper towel.
- Hold the straw upright, with the cotton end at the bottom and roll the bottom end between the thumb and the fore finger in order to free the cotton wool stopper.
- Tap the straw gently to allow air bubble to rise to the top.
- Place the straw into the AI gun – cotton end first and cut the sealed end with scissors.
- Properly fix the sheath with the gun using the O-ring

B. **Steps in insemination process**

- A gentle pat on the rump or a soft-spoken word as you approach for insemination, will help to avoid thrilling or exciting the animal.
- Raise the tail with your right hand and gently massage the rectum with the lubricated glove on your left hand. Place the tail on the back side of your left forearm so it will not interfere with the insemination process.
- The cervix “landmark” simulation for rectal palpation and alignment of in-

semination gun (Fig 11. a, b, c)

- Cup your fingers together in a pointed fashion and bulge the vulva slightly
- Gently wipe the vulva with a paper towel to remove excess manure and debris. Be careful not to apply excessive pressure, which may smear or push manure into the vulva and vagina.
- With your left hand make a fist and press down directly on top of the vulva. This will spread the vulva lips allowing clear access to insert the gun tip several inches into the vagina before contacting the vaginal walls.
- Insert the gun at a 45º upward angle to avoid entering the urethral opening and bladder located on the floor of the vagina.
- With the gun about 6 to 8 inches inside the vagina (varying for individuals), raise the rear of the gun to a somewhat level position and slide it forward until it contacts the external portion of the cervix. You will note a distinct gristly sensation on the gun when it contacts the end of the cervix as the opening into the cervix protrudes back into the vagina, called the fornix.
- Inert the insemination gun in to the cervix located on the floor of the pelvic cavity or near the anterior end of the pelvic bone.
- Use your palm and these two fingers to guide the gun tip to the cervical opening located between your thumb and forefingers. Place the cervix on or over the insemination gun, guide the tip of the insemination gun to dispose the semen in uterine body (Fig 12.1a, b, c).
- Raise your finger and push the plunger slowly so that drops of semen fall directly into the uterine body (Fig 13. a, b, c); don't inseminate in the horns. Make sure you push in with the plunger and do not pull back on the gun, deposit semen in uterine body, and slowly pull the gun from the reproductive tract.
- Remove the gloved hand from the rectum and shake off the excess manure. Check the gun tip for signs of blood, infection or semen leakage inside the sheath.
- Remove the sheath from the gun, pull out the emptied straw and check to

confirm which bull you have used. Dispose of the used glove and sheath in a proper receptacle.

- Wipe the gun clean and dry and return it to the proper storage location.
- Make notes for your veterinarian or future reference where appropriate.

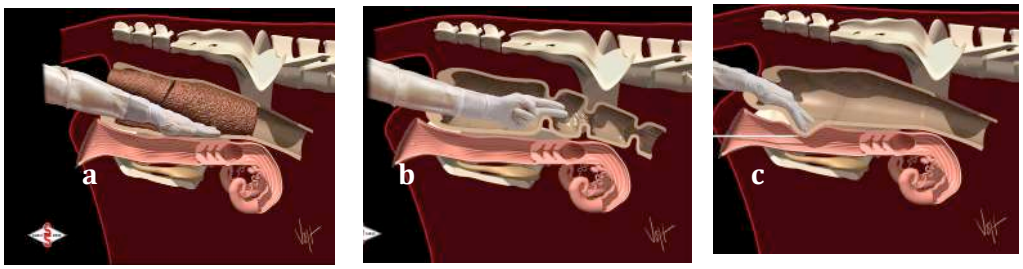


Fig 11. a, b, c. Simulation of left hand manipulation cervix during Recto vaginal insemination

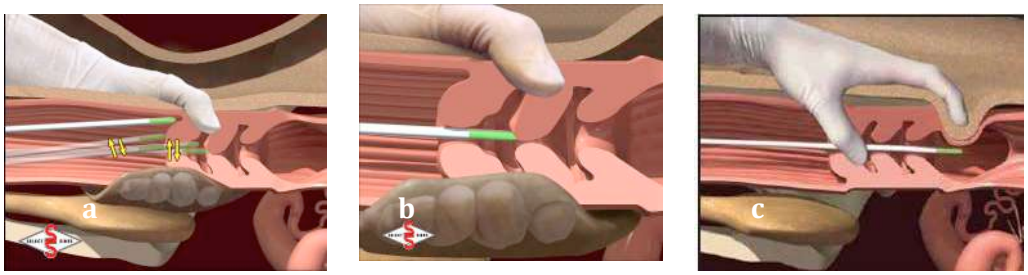


Fig 12. a, b, c Simulation of left hand manipulation cervis to insert insemination gun during Recto vaginal insemination

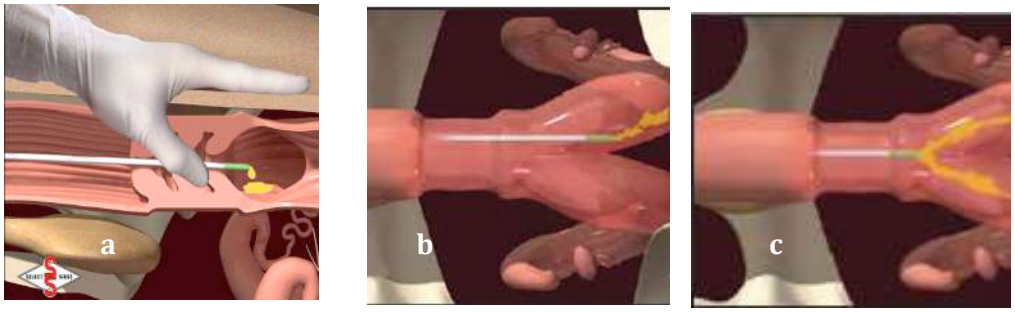


Fig 13. Simulation for the correct (a) and incorrect semen deposition site in the uterus during (b,c) Recto vaginal insemination

Annex 2.

Sub clinical mastitis screening methods and test procedures

1. Screening Test Procedures

- Milk and discard the first three spades of milk from all four quarters on strip cup to visualize the flakes and clot in mastitic milk on black background (Fig 14a)
- Again, take milk from each quarter into different CMT rack quarters and mark with its respective quarters.
- Mix equal amount of CMT reagent with milk samples to each rack quarters (Fig 14b).
- Stir gently and observe gel formation.
- Appreciate more viscosity of the milk that indicates more dead cells and infection.



Figure 14. Screening test methods for subclinical mastitis a. Strip cup test; b. CMT

2. Mastitis Prevention and Control

- Carry out complete milking of affected cows
- Clean and disinfect teats of affected cows after milking
- Apply intramammary infusions and drugs for the treatment of mastitis. In case of systemic infection, use antibiotic drugs or anti-inflammatory drugs reduce inflammation and pain.
- Use teat dipping (Fig 15), sanitation of teats before milking during dry period

- Milk healthy cows first and chronically affected ones at the end.
- Isolate affected cows from rest of the herd
- Screen regularly for sub clinical mastitis and treat positive animals

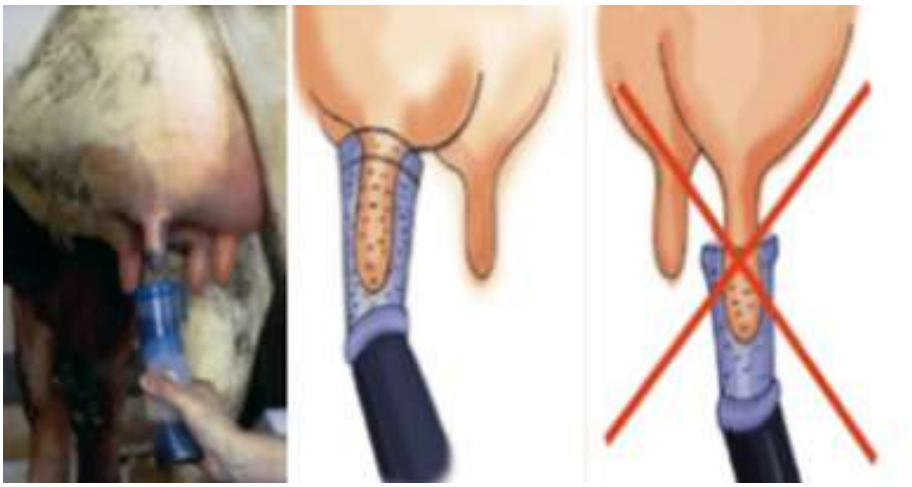


Figure 15. Dipping techniques after milking

Annex 3

Procedures and steps to re-place the prolapsed uterus

- Place the prolapsed mass gently over a clean surface and protect it from soiling
- Do not attempt to remove anything or push the uterus back forcefully, this may cause severe bleeding.
- Gently wash with saline solution to remove any dirt before trying to place back
- Apply dehydrating sugars if available to reduce the swelling of the uterus
- Systematically start inserting the prolapsed uterus from the center and gently push to install it properly
- Apply diluted antibiotic after the whole uterus is inserted fully in place.
- To treat RFM - Use drugs such as immediate post-partum administration of PGF2alpha or oxytocin; Administer intrauterine antibiotic lavage or systemic injection and supportive therapies.

Annex 4.

Neonate calf, pre weaning calf, growing heifers, heifers and breeding cow/heifers feeding regime

Understanding impacts of nutritional management on replacement heifer development is essential to raising heifers that are efficient and profitable. That feed or feed mixture containing all the essential components of a balanced ration which has the ability to fulfill all the needs of a particular animal when given according to its body weight would enable to utilize the genetic potential of dairy cows. Accordingly, as a rule of thumb, the feeding management during this period of the calf growth should enable weaning at 3 months at approximately 80 kg body weight. The detail feeding of dairy cattle from calf hood to lactation is indicated below.

Colostrum feeding (1-4 days)

The immune system of newborn calves is not completely developed and the immunoglobulins in colostrum contribute a substantial component of the immune protection in newborn calves. Colostrum is the first milk secreted after parturition, contains a higher amount of important nutritious diet and vitamins which are essential to combat disease, is essential to provide passive immunity to newborn calves.

- Make sure that the colostrum is available. Check if the dam is producing sufficient colostrum by test-milking all teats.
- Feed neonate calf within 30 minutes to one hour after birth. Colostrum is critical to the survival of the newborn calf.
- Make sure the calf suckles the dam immediately after birth or feed manually if the calf can't suckle the dam;
- In the feeding management of newborn calves, attention should be given to the time, quantity, and quality of colostrum.

- Calves should receive 5 to 6% of their body weight (1.5 to 1.8 liters for a 30-kilo calf) as colostrum within the first 6 hours and again within the subsequent 6 hours (Table 15).
- If the cow's production of colostrum is insufficient, colostrum from other cows or replacer colostrum supplements may be necessary.
- Ideally, colostrum should be collected from cows within 24 hours of calving and fed fresh. Colostrum can also be collected, frozen, and used later.
- When collecting colostrum, consider freezing it in "serving" portions, or about one liter per container. Once the frozen colostrum has been thawed (warmed back) it should not be re-frozen.
- Frozen colostrum should be slowly warmed in warm water to a final temp of 40-45 degrees centigrade. The colostrum should be stirred frequently during thawing.
- In the absence of colostrum from the natural or foster mother, prepare artificial colostrum as a last resort= one egg (protein source) + a half liter of fresh warm water + half a liter of whole milk (source of lactose and milk protein) + one teaspoonful of oil (energy).
- Calves must be provided with adequate clean drinking water at all times
- If colostrum is bottle-fed, don't use a nipple that does not match the calf's ability to swallow. Especially the first feeding of colostrum can cause problems if the feeding rate is faster than the swallowing rate.
- Don't protect the calf from suckling the colostrum milk specially for 3-4 days after parturition.
- Animal colostrum for human consumption is not advisable. It may create scarcity and avoid human competition with the newborn if the practice is common in the area.

Table15. Amount of colostrum and milk to be fed to growing calves

Age of calf (days)	Number of days	Amount in litter	
		Morning	After noon
4	4 (Ccolostrum)	2	2
15	11	1.5	1.5
42	27	2	2
63	21	1.5	1.5
84	21	1	1.1
98	14	0.5	0.5

Milk replacers

These are commercial products manufactured to resemble milk and are mostly used when there is no milk to feed the calf e.g where a cow is sick or died during calving. They are also used when demand and price of milk is high. Preserved colostrums should be used as much as possible before a farmer decides to use milk replacer. The milk replacer should contain 22% protein (if all protein is from milk sources) or 24% when some plant protein is included (on DM basis). Milk replacers are always of lower quality than whole milk and should only be fed if they are cheaper.

Pre-starter

A pre-starter is a high-quality calf feed, which should be low in fiber and is almost similar to milk replacer and is usually fed during the second and third week. It is fed in a dry pelleted form or as a meal. It should be used early to stimulate calves to eat dry feed to enhance rumen development. It is estimated that it takes rumen growth about three weeks after the calf starts eating a handful of dry feed, thus the earlier they start the better. The starter contains slightly higher fiber content compared with the milk replacer. At this stage the calf is consuming little milk and is in transition to becoming a ruminant.

Roughage

Calves should be offered only high-quality forages early in life and supplemented with concentrates (calf starter) (Table 16). If hay is used, it should be of high quality, fine texture, mixed with legumes and fed ad lib. If they are on pasture, it would be best to always graze calves ahead of adults to control parasites.

Water

Calves should be offered fresh water in addition to milk. Lack of drinking water slows down digestion and development of the rumen, and hence the longer it takes before calves can be safely weaned. Between three weeks and weaning, calves' water consumption usually increases and should be available all the time.

Table 16. Example of a feeding schedule for calves

Age of calf (days)	Milk kg/day	Total Milk (kg)	Calf starter (kg/d)	Roughage
1 to 7	Colostrum			
8 to 21	5	70	Handful	
22 to 42	6	126	0.5	Yes (ad lib)
43 to 56	5	70	0.5	Yes (ad lib)
57 to 63	4	28	1	Yes (ad lib)
64 to 77	3	42	1	Yes (ad lib)
78 to 84	2	14	1.5	Yes (ad lib)
Wean the calf				
Total		350	55	

Early weaning feeding management of a calf

Early weaning (5 to <8 weeks) may be adopted to reduce the milk feeding period and labour required for calf rearing. Early weaning require a specific feeding program using low levels of milk and high energy, high protein concentrates, preferably pelleted to stimulate rumen development. Liquid milk or milk replacer is reduced from 3 weeks of age to encourage the calf to consume and maximize intake of dry feeds (Table 17).

Local experience tells that a calf can be weaned at the age of 70 days when it is being fed with a milk partially replaced with whey.

Table 17. Partial replacement of milk for whey for Early weaning of a calf

Calf's age (days)	Daily consumption (liters)		
	Colostrum	Milk	Whey
1-4	3.0		
5-15	-	3.0	-
16-20	-	4.0	-
21-42	-	2.0	2.5
43-63	-	1.5	1.5
64-70	-	1.0	1.0
Total consumption	-	124	

Additionally, the calf would need to be provided with good quality roughage and water (ad lib); and calf starter or formulated concentrate (> 20% CP) as indicated the previous table

Heifers feeding management

- Mature heifers include the age between one year to first calving
- Replacement heifer diets should be formulated with adequate Metabolizable energy and protein to meet the demand for lean growth. Excess energy intake should be avoided pre-pubertal for replacement heifer to reduce adipose tissue deposition.
- The heifers feed options should be based on a concentrate-to-roughage ratio of 50:50 that can be used to support the optimum growth of cross-bred dairy heifers. When feeding heifers, the farmer should aim to achieve a growth rate of 0.5-0.7 kg/d and first calving at 22 to 24 months of age. Mature heifers reach target live weights (230 kg) for breeding at 14-16 months of age (Table 18).
- If allowed to graze good quality pasture alone, it can fulfill the nutrient requirements of heifers (growth and maintenance). In the absence this supplementation with concentrate should be made at 1% of their body weight.

- Drinking water and mineral salt supplement is recommended on a free-choice basis.

Table 18. Preparation of supplemental concentrate mix for young growing heifers after weaning

Option-1	Option-2
✓ 50 kg = wheat bran/corn flour ✓ 48 kg = oilseed cakes ✓ 1 kg = salt ✓ 1 kg = Di-calcium phosphate	✓ 50 shovel = wheat bran/corn flour ✓ 48 shovel oil seed cakes ✓ Half a shovel/jog full of salt ✓ Half a jog/shovel full of Di-calcium phosphate
Total = 100 kg	Total = 100 shovel

Note: - If molasses is available, mix one hand in three hands with water and mix it with the roughage feed or supplemental feed; The ration is expected to have ≥ 20% CP and ≥10 MJ, ME/Kg DM; Ca : P shouldn't exceed 1-2 : 1.

- Pregnant heifers may be maintained on good quality forage alone but concentrates should be given if the forage is of low quality. During the last two months of pregnancy, the feeding regime can affect milk production and on-set of heat during the first lactation. A farmer can, therefore, easily adopt the following feeding management (Table 19) once he ensures ad libtum feeding on the roughage diets (well managed pasture grazing), clean drinking water and mineral block leak.

Table 19. Daily concentrate allowance (post weaning feeding management to calving)

S/N	Calf's age (months)	Concentrate allowance (kg/day/head)
1	13 month up until 7th month of the pregnancy	2 kg
2	8 weeks before calving	3 kg (depends on roughage quality)
3	Last 4 weeks before delivery	4 kg (depends on roughage quality)

Note that it would be sufficient to provide 1-2 kg/day of a concentrate (>18% CP) for calf maintained on good quality forage/well managed grazing pasture.

Feeding Management of Lactating Crossbred Cows

- Roughages, water and mineral block leak shall be given ad libtum while dairy formula feeds (formulated concentrate mix) are offered at the rate of 0.5 kg per each liter of milk production during early lactation (72 to 80 days) and 0.3 kg per each liter of milk production during mid and late lactations (80 to 305 days) (Table 20). However, exceptions that such recommendation on the daily concentrate allowance will vary with the quality of the basal diet; flat rates such as 0.5 per kg per milk production is often being recommended across the entire lactation periods (early to late lactation) for cows maintained on very poor-quality basal diets including crop residues.
- Typical examples of lactating crossbred cows' rations are indicated in table 20 below.

The supplemental feed can be easily prepared from the following feed ingredients.

Table 20. Concentrate diet formulated for lactating cows under local conditions

Option-1	Option-2
✓ 68 kg = wheat bran/corn flour ✓ 30 kg = oil seed cakes ✓ 1 kg = salt ✓ 1 kg di-calcium phosphate	✓ Two-thirds hand/shovel=wheat bran/corn flour ✓ One-third hand/shovel oil seed cakes/fagulo ✓ Half jog full of salt ✓ Half jog full of di-calcium phosphate
Total = 100 kg	Total = One quintal

The concentrate is expected to have 18% CP and 12 MJ, ME/kg DM. Ratio of Ca : P shouldn't exceed 1-2 : 1.